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Chemical Kinetics and Nuclear Chemistry

JEE Main Chemistry · Previous Year Questions with Answer Key

243	200	2002 – 2026	2	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	136 questions
Numerical	105 questions
Out of syllabus	2 questions
Answer key	at the end

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2 **Check**
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MCQ (Single Correct Answer) 136 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

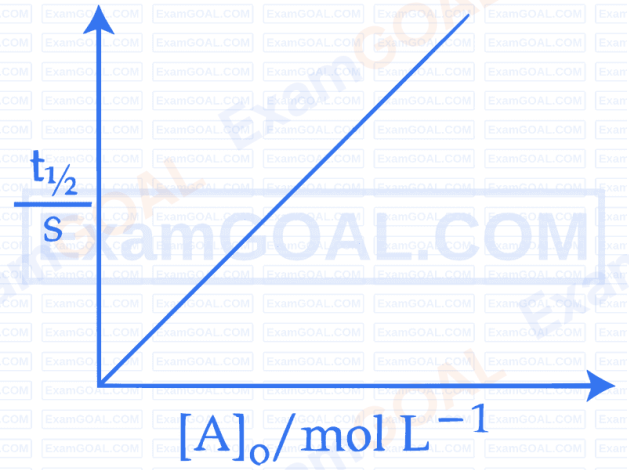
Open solution

Given below are two statements :

$\mathrm{R} = 8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$ and $1\mathrm{cal} = 4.2\,\mathrm{J}$

Statement I : When $\mathrm{Ea} = 12.6\mathrm{kcal/mol}$, the room temperature rate constant is doubled by a $10^{\circ}\mathrm{C}$ increase in temperature (298 K to 308 K)

Statement II : For a first order reaction $\mathrm{A} \rightarrow \mathrm{B}$,



Here $[A]_0$ is the initial concentration of A and $t_{1/2}$ is half life of reaction.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q2. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

First order gas phase reaction

$\mathrm{A} \rightarrow \mathrm{B} + \mathrm{C}$

$p_i =$ initial pressure of gas A, $p_t =$ total pressure of the reaction mixture at time t

Expression of rate constant (k) is

(A) $\frac{1}{t}\ln\frac{p_i}{2p_i - p_t}$

(B) $\frac{1}{t}\ln\frac{2p_i}{p_i - p_t}$

(C) $\frac{1}{t}\ln\frac{p_i}{3p_i - 2p_t}$

(D) $\frac{1}{t}\ln\frac{3p_i}{4p_i - p_t}$

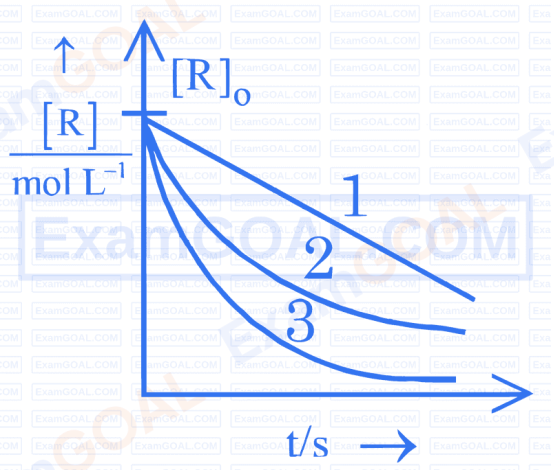
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Q3. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Consider the given graph showing variation of reactant concentration with time.

Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct?



(A) The order of all the three reactions is same.

(B) The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

(C) The SI unit of rate constant of reaction 1 is s^{-1}.

(D) Thermal decomposition of HI on gold surface is an example of reaction 2 .

Q4. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Consider the first order reaction $R \rightarrow P$.

The fraction of molecules decomposed in the given first order reaction can be expressed as

(A) $1 - e^{k_1 t}$ (B) $1 + e^{k_1 t}$ (C) $1 + e^{-k_1 t}$ (D) $1 - e^{-k_1 t}$

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Q5. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

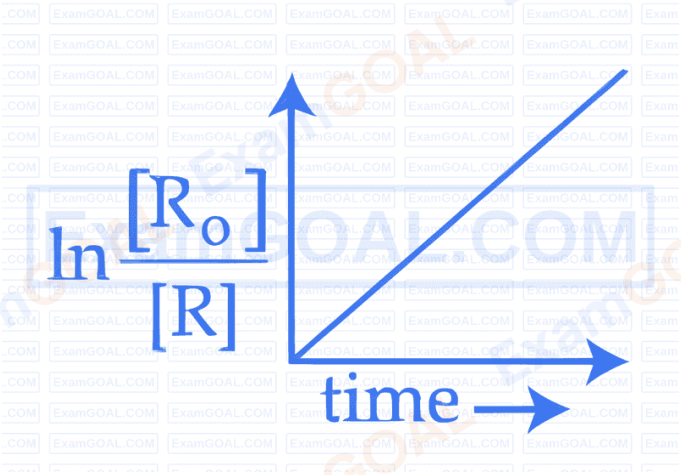
Consider the reaction aX → bY, for which the rate constant at 30°C is 1×10^{-3} mol^{-1} L s^{-1}. Which of the following statements are true?

A. When concentration of 'X' is increased to four times, the rate of reaction becomes 16 times.

B. The reaction is a second order reaction.

C. The half-life period is independent of the concentration of X.

D. Decomposition of N_2O_5 is an example of the above reaction.

E. 

vs time is valid for the above reaction.

Choose the **correct** answer from the options given below:

(A) A and B Only

(B) A, B and C Only

(C) A, B, D and E Only

(D) C and D Only

Q6. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

$\mathbf{t_{100\%}}$ is the time required for the 100% completion of the reaction while $\mathbf{t_{1/2}}$ is the time required for 50% of the reaction to be completed. Which of the following option correctly represents the relation between $t_{100\%}$ and $t_{1/2}$ for zero and first order reactions respectively?

(A) $t_{100\%} = (t_{1/2})^2$ and $t_{100\%} = (t_{1/2})^{-\infty}$

(B) $t_{100\%} = 2t_{1/2}$ and $t_{100\%} = (t_{1/2})^{\infty}$

(C) $t_{100\%} = 2t_{1/2}$ and $t_{100\%} = (2t_{1/2})^2$

(D) $t_{100\%} = (t_{1/2})^{\infty}$ and $t_{100\%} = 2t_{1/2}$

Q7. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{\text{th}}$ and $\left(\frac{1}{10}\right)^{\text{th}}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively.

What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

$(\log 2 = 0.3)$

(A) 30

(B) 9

(C) 3

(D) 0.9

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Q8. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

$\mathrm{A} \rightarrow \mathrm{D}$ is an endothermic reaction occurring in three steps (elementary).

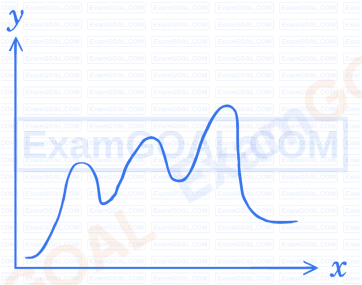
(i) $\mathrm{A} \rightarrow \mathrm{B} \Delta \mathrm{H}_i = +\mathrm{ve}$

(ii) $\mathrm{B} \rightarrow \mathrm{C} \Delta \mathrm{H}_{ii} = -\mathrm{ve}$

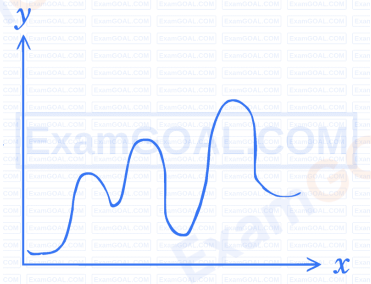
(iii) $\mathrm{C} \rightarrow \mathrm{D} \Delta \mathrm{H}_{iii} = -\mathrm{ve}$

Which of the following graphs between potential energy (y-axis) vs reaction coordinate (x-axis) correctly represents the reaction profile of $A \rightarrow D$?

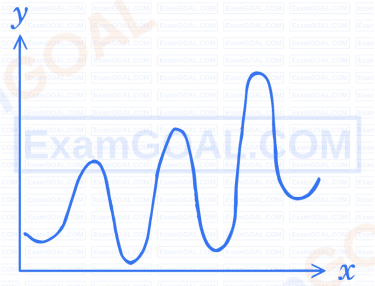
(A)



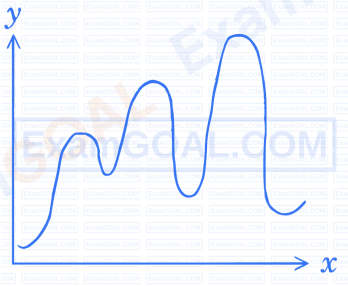
(B)



(C)



(D)



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Q9. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

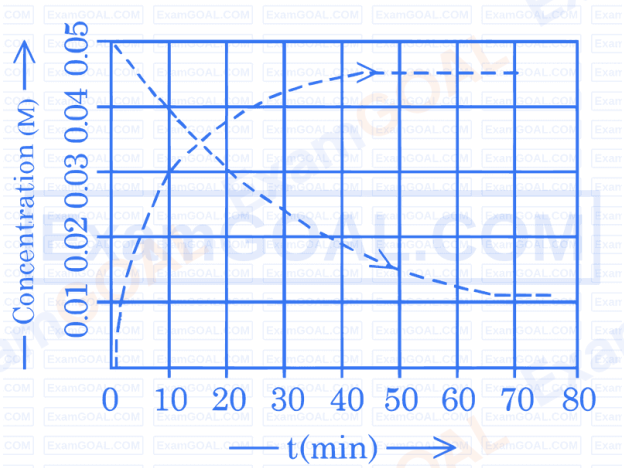
At $27^{\circ}\mathrm{C}$ in presence of a catalyst, activation energy of a reaction is lowered by $10\ \mathrm{kJ\ mol^{-1}}$. The logarithm of ratio of $\frac{\text{k (catalysed)}}{\text{k (uncatalysed)}}$ is....

(Consider that the frequency factor for both the reactions is same)

(A) 1.741 (B) 0.1741 (C) 17.41 (D) 3.482

Q10. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution



Given above is the concentration vs time plot for a dissociation reaction : $\mathrm{A} \rightarrow \mathrm{nB}$.

Based on the data of the initial phase of the reaction (initial 10 min), the value of n is ____ .

(A) 2 (B) 5 (C) 4 (D) 3

Q11. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

Observe the following reactions at $\mathrm{T(K)}$.

I. $\mathrm{A} \rightarrow$ products.

II. $5\mathrm{Br^-(aq)} + \mathrm{BrO_3^-(aq)} + 6\mathrm{H^+(aq)} \rightarrow 3\mathrm{Br_2(aq)} + 3\mathrm{H_2O(l)}$

Both the reactions are started at 10.00 am . The rates of these reactions at 10.10 am are same. The value of $-\frac{\Delta[\mathrm{Br^-}]}{\Delta t}$ at 10.10 am is $2 \times 10^{-4}\ \mathrm{mol\ L^{-1}\ min^{-1}}$. The concentration of A at 10.10 am is $10^{-2}\ \mathrm{mol\ L^{-1}}$. What is the first order rate constant (in $\mathrm{min^{-1}}$) of reaction I ?

(A) 4×10^{-3} (B) 2×10^{-3} (C) 10^{-3} (D) 10^{-2}

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Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Correct statements regarding Arrhenius equation among the following are :

A. Factor $e^{-Ea/RT}$ corresponds to fraction of molecules having kinetic energy less than Ea.

B. At a given temperature, lower the Ea, faster is the reaction.

C. Increase in temperature by about $10^{\circ}C$ doubles the rate of reaction.

D. Plot of $\log k$ vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{Ea}{R}$.

Choose the correct answer from the options given below :

(A) A and B Only (B) B and D Only (C) B and C Only (D) A and C Only

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

$A \rightarrow$ product (First order reaction).

Three sets of experiment were performed for a reaction under similar experimental conditions:

Run $1 \Rightarrow 100\ mL$ of 10 M solution of reactant A

Run $2 \Rightarrow 200\ mL$ of 10 M solution of reactant A

Run $3 \Rightarrow 100\ mL$ of 10 M solution of reactant $A + 100\ mL$ of H_2O added.

The correct variation of rate of reaction is

(A) Run $1 =$ Run $2 =$ Run 3 (B) Run $3 <$ Run $1 <$ Run 2

(C) Run $1 <$ Run $2 <$ Run 3 (D) Run $3 <$ Run $1 =$ Run 2

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Decomposition of A is a first order reaction at T(K) and is given by **A(g) → B(g) + C(g).**

In a closed 1 L vessel, 1 bar A(g) is allowed to decompose at T(K). After 100 minutes, the total pressure was 1.5 bar. What is the rate constant (in min^{-1}) of the reaction? (log 2 = 0.3)

(A) 6.9×10^{-4} (B) 6.9×10^{-3} (C) 6.9×10^{-1} (D) 6.9×10^{-2}

Q15. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one eighth of its initial concentration are t_1 and t_2 (s), respectively. The ratio t_1/t_2 will be:

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

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Q16. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

$A(g) \rightarrow B(g) + C(g)$ is a first order reaction.

Time	t	∞
P_{system}	P_t	P_∞

The reaction was started with reactant A only. Which of the following expressions is correct for rate constant k?

(A) $k = \frac{1}{t} \ln \frac{p_\infty}{p_t}$

(B) $k = \frac{1}{t} \ln \frac{p_\infty}{2(p_\infty - p_t)}$

(C) $k = \frac{1}{t} \ln \frac{2(p_\infty - p_t)}{p_t}$

(D) $k = \frac{1}{t} \ln \frac{p_\infty}{(p_\infty - p_t)}$

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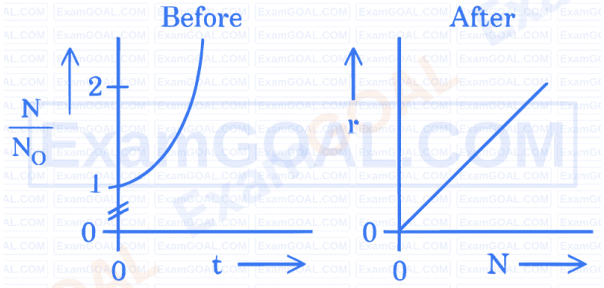
Q17. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

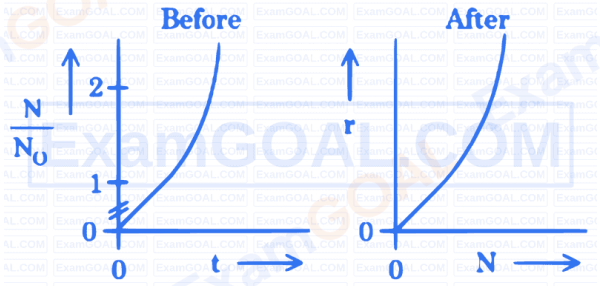
A person's wound was exposed to some bacteria and then bacterial growth started to happen at the same place. The wound was later treated with some antibacterial medicine and the rate of bacterial decay(r) was found to be proportional with the square of the existing number of bacteria at any instance. Which of the following set of graphs correctly represents the 'before' and 'after' situation of the application of the medicine?

[Given: $N =$ No. of bacteria, $t =$ time, bacterial growth follows 1st order kinetics.]

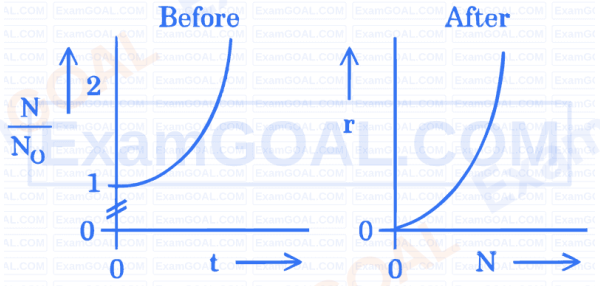
(A)



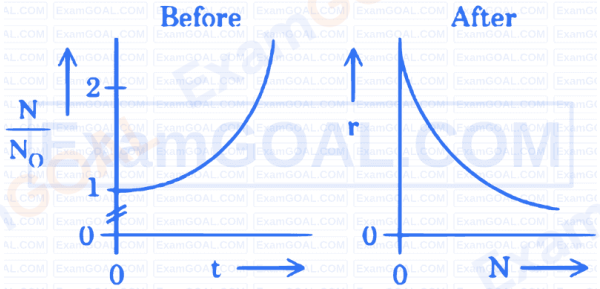
(B)



(C)



(D)



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Q18. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Reaction $\mathrm{A(g)} \rightarrow 2\,\mathrm{B(\,g)} + \mathrm{C(g)}$ is a first order reaction. It was started with pure A

t/min	Pressure of system at time t/mm Hg
10	160
∞	240

Which of the following option is incorrect?

(A) Initial pressure of A is 80 mm Hg

(B) The reaction never goes to completion

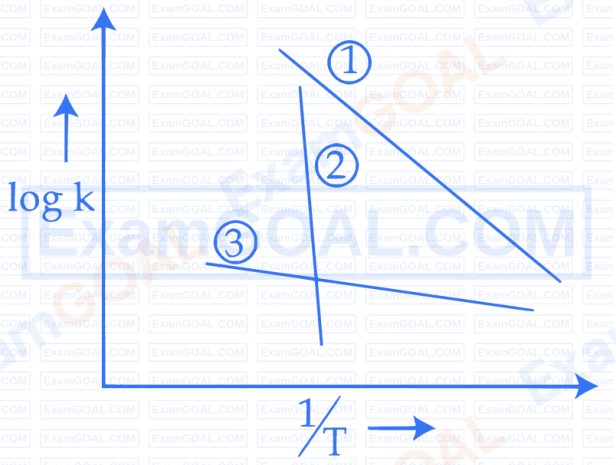
(C) Partial pressure of A after 10 minute is 40 mm Hg

(D) Rate constant of the reaction is $1.693\ \mathrm{min}^{-1}$

Q19. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider the following plots of $\log$ of rate constant $\mathrm{k}(\log \mathrm{k})$ vs $\frac{1}{\mathrm{T}}$ for three different reactions. The correct order of activation energies of these reactions is :



(A) $\mathrm{Ea}_2 > \mathrm{Ea}_1 > \mathrm{Ea}_3$

(B) $\mathrm{Ea}_1 > \mathrm{Ea}_3 > \mathrm{Ea}_2$

(C) $\mathrm{Ea}_3 > \mathrm{Ea}_2 > \mathrm{Ea}_1$

(D) $\mathrm{Ea}_1 > \mathrm{Ea}_2 > \mathrm{Ea}_3$

Q20. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Half life of zero order reaction $\mathrm{A} \rightarrow$ product is 1 hour, when initial concentration of reactant is $2.0\ \mathrm{mol\,L}^{-1}$. The time required to decrease concentration of A from 0.50 to $0.25\ \mathrm{mol\,L}^{-1}$ is :

(A) 0.5 hour (B) 15 min (C) 60 min (D) 4 hour

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Q21. JEE Main 2025 (Online) 4th April Morning Shift

For $A_2 + B_2 \rightleftharpoons 2AB$

E_a for forward and backward reaction are 180 and $200\ kJ\ mol^{-1}$ respectively

If catalyst lowers E_a for both reactions by $100\ kJ\ mol^{-1}$.

Which of the following statement is correct?

(A) Catalyst does not alter the Gibbs energy change of a reaction.

(B) The enthalpy change for the reaction is $+20\ kJ\ mol^{-1}$.

(C) Catalyst can cause non-spontaneous reactions to occur.

(D) The enthalpy change for the catalysed reaction is different from that of uncatalysed reaction.

Open solution

Q22. JEE Main 2025 (Online) 4th April Morning Shift

Rate law for a reaction between A and B is given by

$r = k[A]^n[B]^m$

If concentration of A is doubled and concentration of B is halved from their initial value, the ratio of new rate of reaction to the initial rate of reaction $\left(\frac{r_2}{r_1}\right)$ is

(A) $(n - m)$ (B) $2^{(n-m)}$ (C) $\frac{1}{2^{m+n}}$ (D) $(m + n)$

Open solution

Q23. JEE Main 2025 (Online) 3rd April Evening Shift

Consider the following statements related to temperature dependence of rate constants.

Identify the correct statements.

A. The Arrhenius equation holds true only for an elementary homogenous reaction.

B. The unit of A is same as that of k in Arrhenius equation.

C. At a given temperature, a low activation energy means a fast reaction.

D. A and Ea as used in Arrhenius equation depend on temperature.

E. When $Ea \gg RT$, A and Ea become interdependent.

Choose the correct answer from the options given below:

(A) A, C and D Only (B) B, D and E Only (C) B and C Only (D) A and B Only

Open solution

Q24. JEE Main 2025 (Online) 3rd April Morning Shift

In a reaction $A + B \rightarrow C$, initial concentrations of A and B are related as $[A]_0 = 8[B]_0$. The half lives of A and B are 10 min and 40 min , respectively. If they start to disappear at the same time, both following first order kinetics, after how much time will the concentration of both the reactants be same?

(A) 20 min (B) 40 min (C) 80 min (D) 60 min

Open solution

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Q25. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Reactant A converts to product D through the given mechanism (with the net evolution of heat):

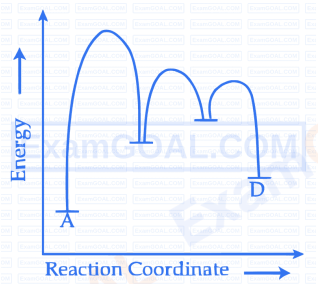
A → B slow; ΔH = +ve

B → C fast; ΔH = -ve

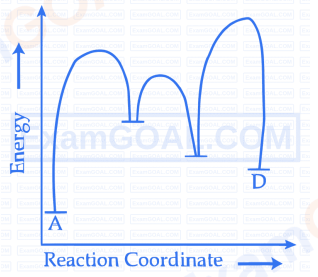
C → D fast; ΔH = -ve

Which of the following represents the above reaction mechanism?

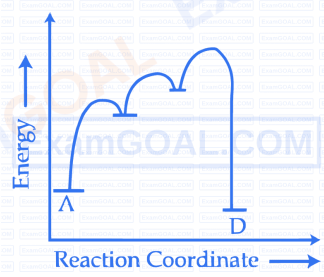
(A)



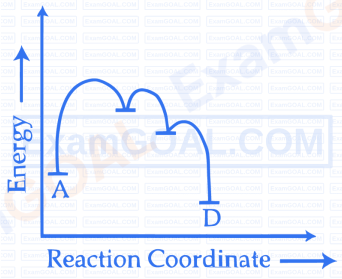
(B)



(C)



(D)



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Q26. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was $16\mathrm{mg/mL}$ which becomes $4\mathrm{mg/mL}$ in 12 months. The expiry time of the drug in months is _____.

Assume that the decomposition of the drug follows first order kinetics.

(A) 12 (B) 3 (C) 6 (D) 2

Q27. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

The reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism:

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A + A$ (fast)

$A + B_2 \xrightarrow{k_2} AB + B$ (slow)

$A + B \rightarrow AB$ (fast)

The overall order of the reaction is:

(A) 3 (B) 2.5 (C) 1.5 (D) 2

Q28. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Consider an elementary reaction

$\mathrm{A(g) + B(g) \rightarrow C(g) + D(g)}$

If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become 'x' times of the original reaction rate. The value of x is :

(A) 3 (B) 9 (C) $\frac{1}{3}$ (D) $\frac{1}{9}$

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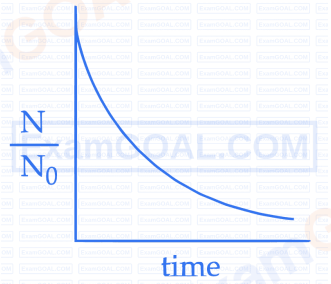
Q29. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

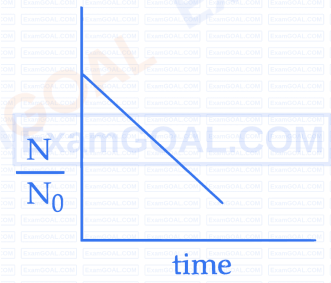
For bacterial growth in a cell culture, growth law is very similar to the law of radioactive decay. Which of the following graphs is most suitable to represent bacterial colony growth ?

Where N - Number of Bacteria at any time, N_0 - Initial number of Bacteria.

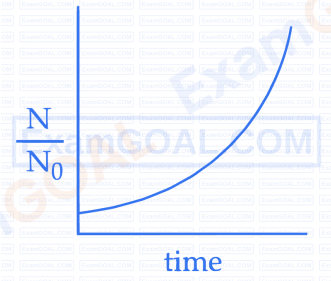
(A)



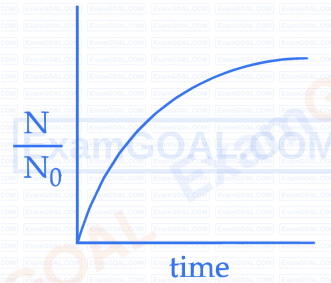
(B)



(C)



(D)



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Q30. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

$[A]_0$ / mol L^{-1}	$t_{1/2}$ / min
0.100	200
0.025	100

For a given reaction $R \rightarrow P$, $t_{1/2}$ is related to $[A]_0$ as given in table.

Given: $\log 2 = 0.30$

Which of the following is true?

A. The order of the reaction is $1/2$.

B. If $[A]_0$ is 1 M , then $t_{1/2}$ is $200\sqrt{10}$ min

C. The order of the reaction changes to 1 if the concentration of reactant changes from 0.100 M to 0.500 M.

D. $t_{1/2}$ is 800 min for $[A]_0 = 1.6M$

Choose the correct answer from the options given below:

(A) A and C Only **(B)** A, B and D Only **(C)** A and B Only **(D)** C and D Only

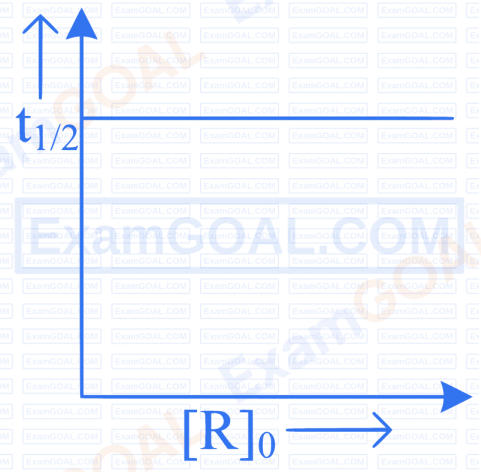
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Q31. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

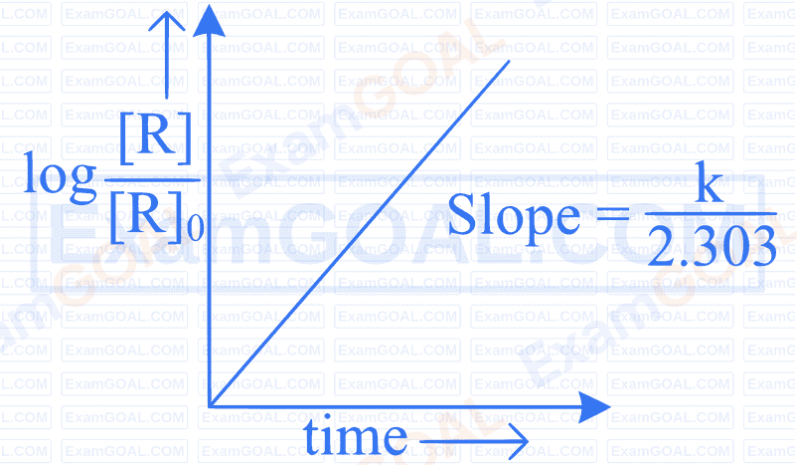
Given below are two statements :

Statement (I) :



is valid for first order reaction.

Statement (II) :



is valid for first order reaction.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

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Q32. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

For a reaction, $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$ in a constant volume container, no products were present initially. The final pressure of the system when 50% of reaction gets completed is

(A) $5/2$ times of initial pressure

(B) $7/2$ times of initial pressure

(C) $7/4$ times of initial pressure

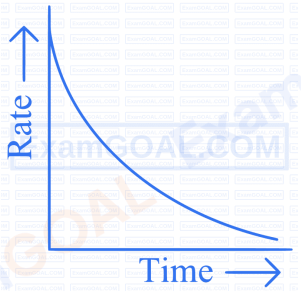
(D) 5 times of initial pressure

Q33. JEE Main 2025 (Online) 23rd January Evening Shift

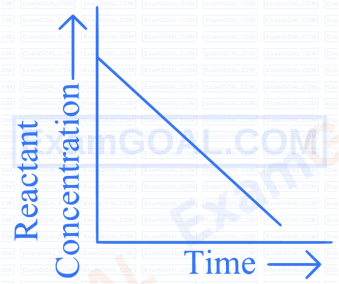
Open solution

Which of the following graphs most appropriately represents a zero order reaction?

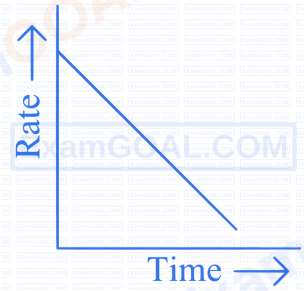
(A)



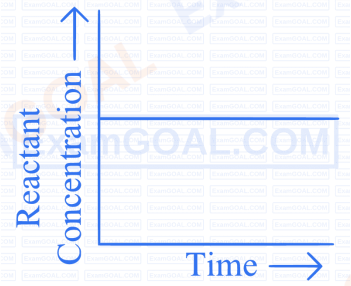
(B)



(C)



(D)

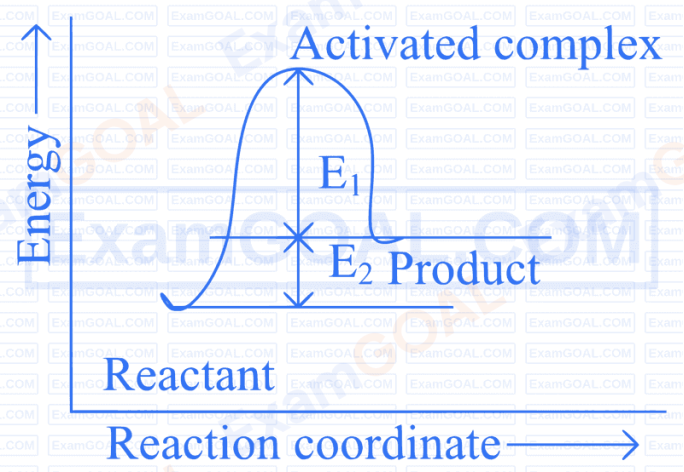


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Q34. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Consider the given figure and choose the correct option :



(A) Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product.

(B) Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant.

(C) Activation energy of backward reaction is E_1 and product is more stable than reactant.

(D) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

Q35. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

For a reaction $A \xrightarrow{K_1} B \xrightarrow{K_2} C$ If the rate of formation of B is set to be zero then the concentration of B is given by :

(A) $(\mathrm{K}_1 - \mathrm{K}_2)[\mathrm{A}]$

(B) $(\mathrm{K}_1 + \mathrm{K}_2)[\mathrm{A}]$

(C) $\mathrm{K}_1\mathrm{K}_2[\mathrm{A}]$

(D) $(\mathrm{K}_1/\mathrm{K}_2)[\mathrm{A}]$

Q36. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Integrated rate law equation for a first order gas phase reaction is given by (where P_i is initial pressure and P_t is total pressure at time t)

(A) $k = \frac{2.303}{t} \times \log \frac{P_i}{(2P_i - P_t)}$

(B) $\mathrm{k} = \frac{2.303}{\mathrm{t}} \times \log \frac{(2\mathrm{P_i} - \mathrm{P_t})}{\mathrm{P_i}}$

(C) $k = \frac{2.303}{t} \times \frac{P_i}{(2P_i - P_t)}$

(D) $\mathrm{k} = \frac{2.303}{\mathrm{t}} \times \log \frac{2\mathrm{P_i}}{(2\mathrm{P_i} - \mathrm{P_t})}$

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Q37. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

For a chemical reaction $A + B \rightarrow$ Product, the order is 1 with respect to A and B.

Rate $mol\ L^{-1}\ S^{-1}$	$[A]$ $mol\ L^{-1}$	$[B]$ $mol\ L^{-1}$
0.10	20	0.5
0.40	x	0.5
0.80	40	y

What is the value of x and y ?

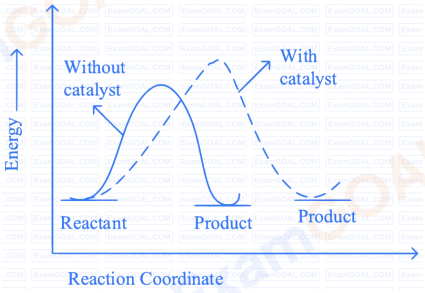
(A) 80 and 4 (B) 160 and 4 (C) 80 and 2 (D) 40 and 4

Q38. JEE Main 2023 (Online) 8th April Evening Shift

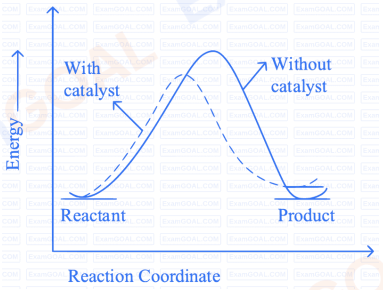
Open solution

The correct reaction profile diagram for a positive catalyst reaction.

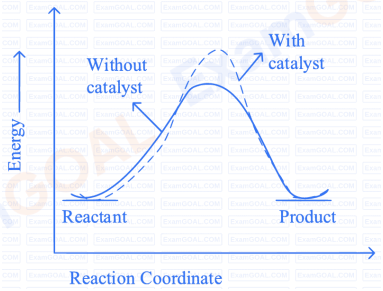
(A)



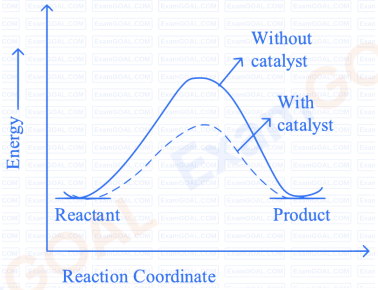
(B)



(C)



(D)

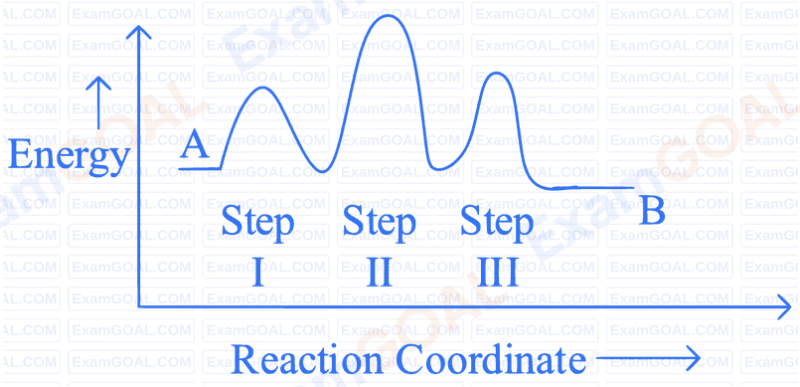


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Q39. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Consider the following reaction that goes from A to B in three steps as shown below:



Choose the correct option

(A)

Number of intermediates	Number of Activated complexes	Rate determining step
2	3	III

(B)

Number of intermediates	Number of Activated complexes	Rate determining step
2	3	II

(C)

Number of intermediates	Number of Activated complexes	Rate determining step
2	3	I

(D)

Number of intermediates	Number of Activated complexes	Rate determining step
3	2	II

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Q40. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

A student has studied the decomposition of a gas AB_3 at 25°C. He obtained the following data.

p (mm Hg)	50	100	200	400
relative $t_{1/2}$ (s)	4	2	1	0.5

The order of the reaction is

(A) 2 **(B)** 0.5 **(C)** 1 **(D)** 0 (zero)

Q41. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :

(A) Always use freshly prepared starch solution.

(B) Always keep the concentration of sodium thiosulphate solution less than that of KI solution.

(C) Record the time immediately after the appearance of blue colour.

(D) Record the time immediately before the appearance of blue colour.

(E) Always keep the concentration of sodium thiosulphate solution more than that of KI solution.

Choose the correct answer from the options given below :

(A) $(A), (B), (C)$ only **(B)** $(A), (D), (E)$ only

(C) $(D), (E)$ only **(D)** $(A), (B), (E)$ only

Q42. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

At $30^\circ C$, the half life for the decomposition of AB_2 is $200\ s$ and is independent of the initial concentration of AB_2. The time required for 80% of the AB_2 to decompose is

Given: $\log 2 = 0.30 \quad \log 3 = 0.48$

(A) 200 s **(B)** 323 s **(C)** 467 s **(D)** 532 s

Q43. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

For a first order reaction, the time required for completion of 90% reaction is 'x' times the half life of the reaction. The value of 'x' is

(Given : ln 10 = 2.303 and log 2 = 0.3010)

(A) 1.12 **(B)** 2.43 **(C)** 3.32 **(D)** 33.31

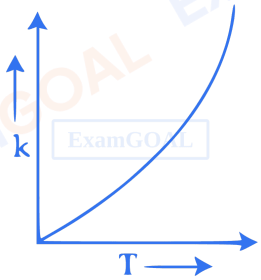
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Q44. JEE Main 2021 (Online) 1st September Evening Shift

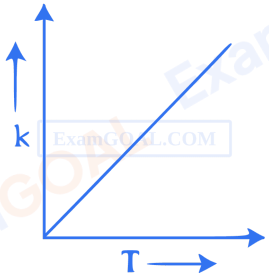
Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction?

Open solution

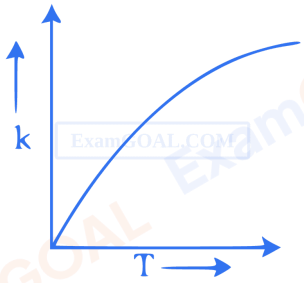
(A)



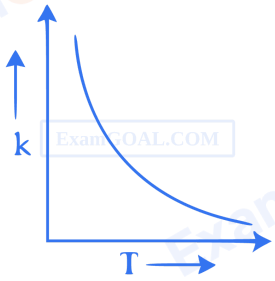
(B)



(C)



(D)



Q45. JEE Main 2021 (Online) 27th July Morning Shift

For a reaction of order n, the unit of the rate constant is :

(A) $mol^{1-n} L^{1-n} s$ (B) $mol^{1-n} L^{2n} s^{-1}$ (C) $mol^{1-n} L^{n-1} s^{-1}$ (D) $mol^{1-n} L^{1-n} s^{-1}$

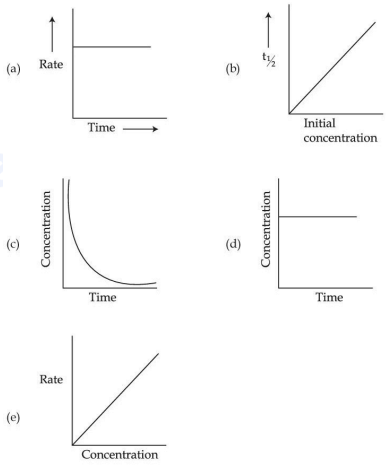
Open solution

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Q46. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

For the following graphs,



Choose from the options given below, the correct one regarding order of reaction is :

(A) (b) Zero order (c) and (e) First order

(B) (a) and (b) Zero order (e) First order

(C) (b) and (d) Zero order (e) First order

(D) (a) and (b) Zero order (c) and (e) First order

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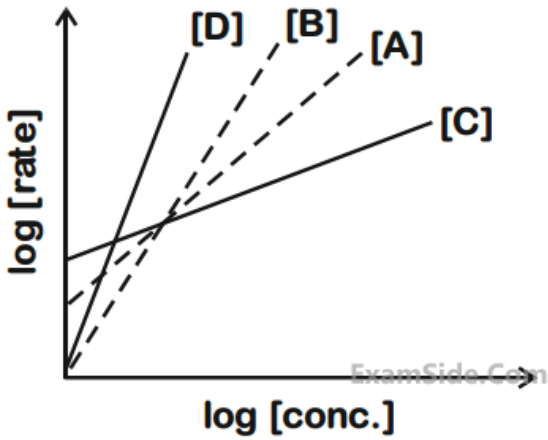
Q47. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

Consider the following reactions

$A \rightarrow P_1$; $B \rightarrow P_2$; $C \rightarrow P_3$; $D \rightarrow P_4$,

The order of the above reactions are a, b, c, and d, respectively. The following graph is obtained when log[rate] vs. log[conc.] are plotted



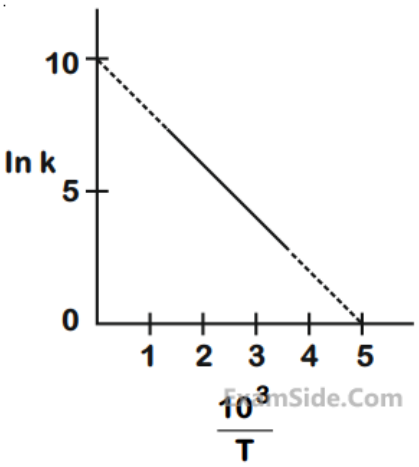
Among the following, the correct sequence for the order of the reactions is :

(A) d > b > a > c (B) d > a > b > c (C) a > b > c > d (D) c > a > b > d

Q48. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

The rate constant (k) of a reaction is measured at different temperatures (T), and the data are plotted in the given figure. The activation energy of the reaction in kJ mol^{-1} is :
(R is gas constant)



(A) R (B) 2R (C) $\frac{1}{R}$ (D) $\frac{1}{2R}$

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Q49. JEE Main 2020 (Online) 5th September Morning Slot

A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B(in s) :
(Use ln 2 = 0.693)

Open solution

(A) 180 (B) 120 (C) 300 (D) 900

Q50. JEE Main 2020 (Online) 3rd September Evening Slot

For the reaction
$2A + 3B + \frac{3}{2}C \rightarrow 3P$, which statement is correct ?

Open solution

(A) $\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$
(B) $\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$
(C) $\frac{dn_A}{dt} = \frac{3}{2}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$
(D) $\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{4}{3}\frac{dn_C}{dt}$

Q51. JEE Main 2020 (Online) 3rd September Morning Slot

It is true that :

Open solution

(A) A first order reaction is always a single step reaction
(B) A zero order reaction is a multistep reaction
(C) A zero order reaction is a single step reaction
(D) A second order reaction is always a multistep reaction

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Q52. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

The results given in the below table were obtained during kinetic studies of the following reaction

$2A + B \rightarrow C + D$

Experiment	$[A]/molL^{-1}$	$[B]/molL^{-1}$	Initial rate/$molL^{-1}$ min^{-1}
I	0.1	0.1	6.00×10^{-3}
II	0.1	0.2	2.40×10^{-2}
III	0.2	0.1	1.20×10^{-2}
IV	X	0.2	7.20×10^{-2}
V	0.3	Y	2.88×10^{-1}

X and Y in the given table are respectively :

(A) 0.3, 0.4 (B) 0.4, 0.3 (C) 0.4, 0.4 (D) 0.3, 0.3

Q53. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

For the following reactions

$A \xrightarrow{700K}$ Product

$A \xrightarrow[catalyst]{500K}$ Product

it was found that E_a is decreased by 30 kJ/mol in the presence of catalyst.

If the rate remains unchanged, the activation energy for catalysed reaction is (Assume pre exponential factor is same):

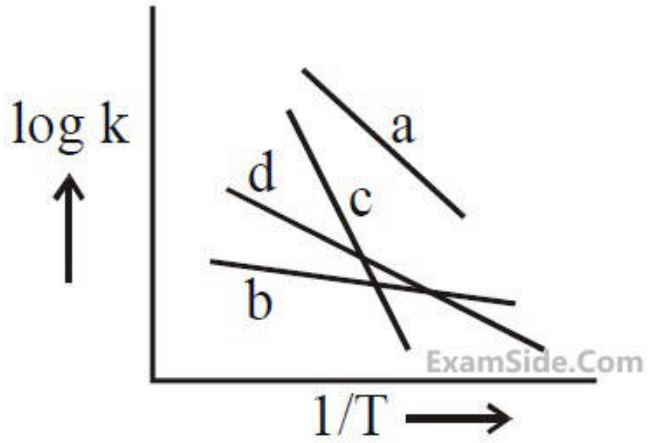
(A) 198 kJ/mol (B) 135 kJ/mol (C) 105 kJ/mol (D) 75 kJ/mol

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Q54. JEE Main 2020 (Online) 8th January Evening Slot

Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the activation energies of these reactions?

Open solution



(A) $E_c > E_a > E_d > E_b$

(B) $E_a > E_c > E_d > E_b$

(C) $E_b > E_a > E_d > E_c$

(D) $E_b > E_d > E_c > E_a$

Q55. JEE Main 2020 (Online) 8th January Morning Slot

The rate of a certain biochemical reaction at physiological temperature (T) occurs 10^6 times faster with enzyme than without. The change in the activation energy upon adding enzyme is :

Open solution

(A) + 6RT

(B) − 6 (2.303)RT

(C) − 6RT

(D) + 6(2.303)RT

Q56. JEE Main 2020 (Online) 7th January Evening Slot

For the reaction

$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$

the observed rate expression is, rate = $K_f[NO]^2[H_2]$. The rate expression for the reverse reaction is :

Open solution

(A) $K_b[N_2][H_2O]$

(B) $K_b[N_2][H_2O]^2/[H_2]$

(C) $K_b[N_2][H_2O]^2/[NO]$

(D) $K_b[N_2][H_2O]^2$

Q57. JEE Main 2019 (Online) 12th April Evening Slot

NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per the equation, $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$.

Open solution

The initial concentration of N_2O_5 is 3.00 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is :

(A) 2.083×10^{-3} mol L^{-1} min^{-1}

(B) 8.333×10^{-3} mol L^{-1} min^{-1}

(C) 4.167×10^{-3} mol L^{-1} min^{-1}

(D) 1.667×10^{-2} mol L^{-1} min^{-1}

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Q58. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

In the following reaction; xA $\rightarrow$ yB

$$\log_{10}\left[-\frac{d[A]}{dt}\right] = \log_{10}\left[\frac{d[B]}{dt}\right] + 0.3010$$

'A' and 'B' respectively can be :

(A) n-Butane and Iso-butane

(B) C_2H_4 and C_4H_8

(C) C_2H_4 and C_6H_6

(D) N_2O_4 and NO_2

Q59. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

For the reaction of H_2 with I_2, the rate constant is 2.5×10^{-4} dm^3 $mol^{-1}s^{-1}$ at 327°C and 1.0 dm^3 $mol^{-1}s^{-1}$ at 527°C. The activation energy for the reaction, in kJ $mole^{-1}$ is : (R = 8.314 JK^{-1} mol^{-1})

(A) 59 (B) 166 (C) 72 (D) 150

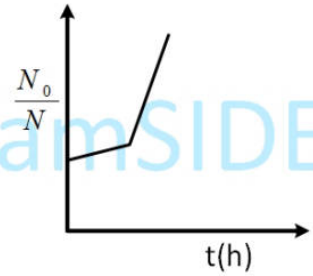
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Q60. JEE Main 2019 (Online) 10th April Morning Slot

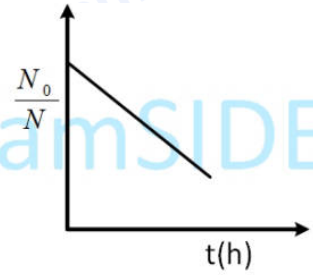
Open solution

A bacterial infection in an internal wound grows as N'(t) = N_0 exp(t), where the time t is in hours. A dose of antibiotic, taken orally, needs 1 hour to reach the wound. Once it reaches there, the bacterial population goes down as $\frac{dN}{dt} = -5N^2$. What will be the plot of $\frac{N_0}{N}$ vs. t after 1 hour?

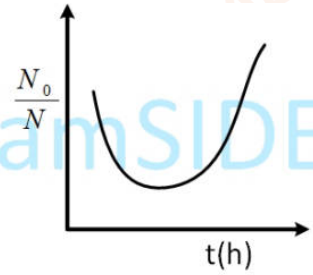
(A)



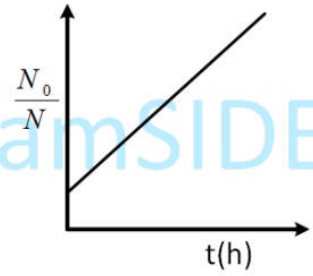
(B)



(C)



(D)



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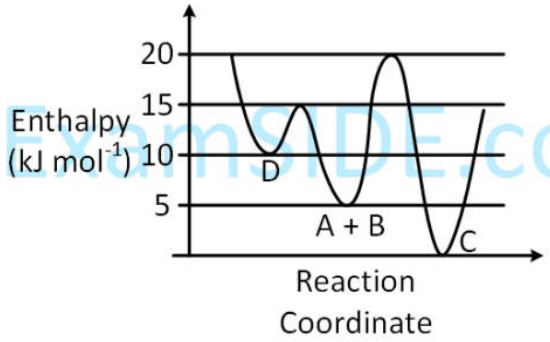
Q61. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Consider the given plot of enthalpy of the following reaction between A and B.

A+ B $\rightarrow$ C + D

Identify the incorrect statement.



(A) Formation of A and B from C has highest enthalpy of activation.

(B) D is kinetically stable product.

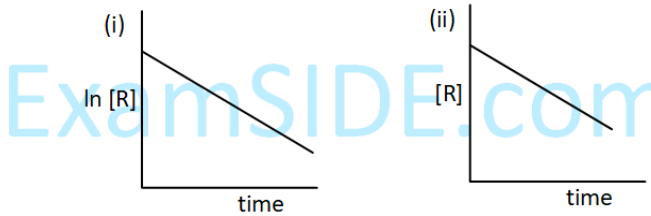
(C) C is the thermodynamically stable product

(D) Activation enthalpy to form C is 5kJ mol^{-1} less than that to form D.

Q62. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The given plots represent the variation of the concentration of a reactant R with time for two different reactions (i) and (ii). The respective orders of the reactions are :



(A) 0,1 (B) 1,0 (C) 0,2 (D) 1,1

Q63. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

For a reaction scheme $A \xrightarrow{k_1} B \xrightarrow{k_2} C$,

if the rate of formation of B is set to be zero then the concentration of B is given by :

(A) $k_1 k_2 [A]$ (B) $\left(\frac{k_1}{k_2}\right)[A]$

(C) $(k_1 + k_2)[A]$ (D) $(k_1 - k_2)[A]$

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Q64. JEE Main 2019 (Online) 8th April Morning Slot

For the reaction 2A + B $\rightarrow$ C, the values of initial rate at diffrent reactant concentrations are given in the table below. The rate law for the reaction is :

Open solution

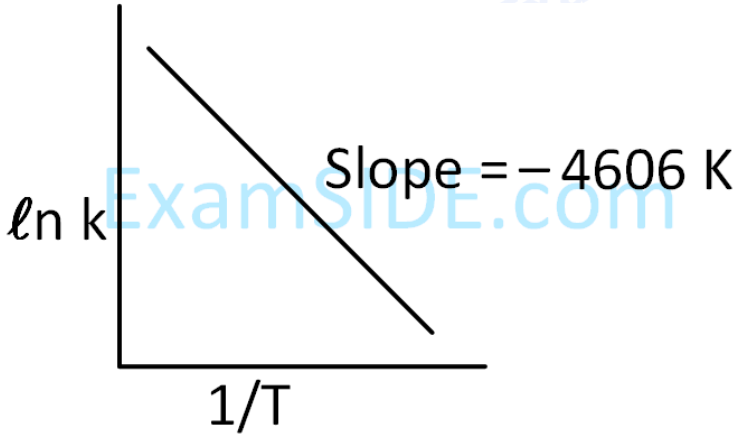
[A] (mol L^{-1})	[B] (mol L^{-1})	Initial Rate (mol $L^{-1}s^{-1}$)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

(A) Rate = $k[A][B]^2$ (B) Rate = $k[A]^2[B]^2$ (C) Rate = $k[A]^2[B]$ (D) Rate = $k[A][B]$

Q65. JEE Main 2019 (Online) 12th January Evening Slot

For a reaction consider the plot of $\ell n\ k$ versus 1/T given in the figure. If the rate constant of this reaction at 400 K is $10^{-5}\ s^{-1}$, then the rate constant at 500 K is –

Open solution



(A) $10^{-4}\ s^{-1}$ (B) $4 \times 10^{-4}\ s^{-1}$ (C) $10^{-6}\ s^{-1}$ (D) $2 \times 10^{-4}\ s^{-1}$

Q66. JEE Main 2019 (Online) 12th January Morning Slot

Decomposition of X exhibits a rate constant of 0.05 μg/year. How many years are required for the decomposition of 5μg of X into 2.5 μg?

Open solution

(A) 50 (B) 20 (C) 25 (D) 40

Q67. JEE Main 2019 (Online) 11th January Evening Slot

The reaction 2X $\rightarrow$ B is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be:

Open solution

(A) 18.0 h (B) 9.0 h (C) 7.2 h (D) 12.0 h

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Q68. JEE Main 2019 (Online) 11th January Morning Slot

If a reaction follows the Arrhenius equation, the plot ln k vs $\frac{1}{(RT)}$ gives straight line with a gradient (– y) unit.

Open solution

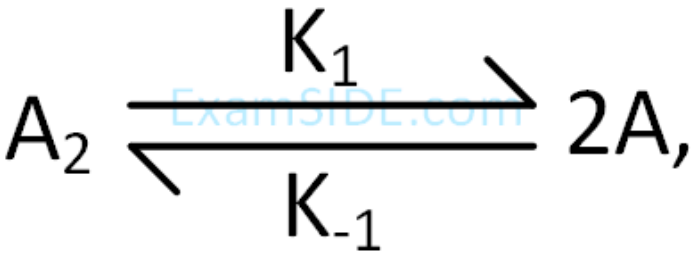
The energy required to active the reactant is :

(A) y unit (B) y/R unit (C) yR unit (D) –y unit

Q69. JEE Main 2019 (Online) 10th January Evening Slot

For an elementary chemical reaction,

Open solution



the expression for $\frac{d[A]}{dt}$ is

(A) $2K_1[A_2] - K_{-1}[A]^2$ (B) $K_1[A_2] - K_{-1}[A]^2$

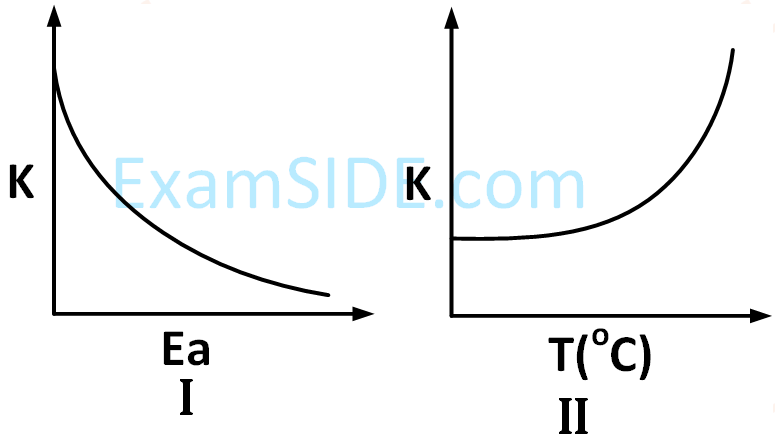
(C) $K_1[A_2] + K_{-1}[A]^2$ (D) $2K_1[A_2] - 2K_{-1}[A]^2$

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Q70. JEE Main 2019 (Online) 10th January Morning Slot

Consider the given plots for a reaction obeying Arrhenius equation ($0^oC < T < 300^oC$) : (K and Ea are rate constant and activation energy, respectively)

Open solution



Choose the correct option :

(A) I is right but II is wrong

(B) Both I and II are correct

(C) Both I and II are wrong

(D) I is wrong but II is right

Q71. JEE Main 2019 (Online) 9th January Evening Slot

For the reaction, $2A + B \rightarrow$ products, when the concentrations of A and B both were doubled, the rate of the reaction increased from 0.3 mol $L^{-1}s^{-1}$ to 2.4 mol $L^{-1}s^{-1}$. When the concentration of A alone is doubled, the rate increased from 0.3 mol $L^{-1}s^{-1}$ to 0.6 mol $L^{-1}s^{-1}$.

Open solution

(A) Total order of the reaction is 4

(B) Order of the reaction with respect to B is 2

(C) Order of the reaction with respect to B is 1

(D) Order of the reaction with respect to A is 2

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Q72. JEE Main 2019 (Online) 9th January Morning Slot

The following results were obtained during kinetic studies of the reaction ;

$2A + B \rightarrow$ Products

Open solution

Experiment	[A] (in mol L^{-1})	[B] (in mol L^{-1})	Initial Rate of reaction (In mol L^{-1} min^{-1})
I	0.10	0.20	6.93×10^{-3}
II	0.10	0.25	6.93×10^{-3}
III	0.20	0.30	1.386×10^{-2}

The time (in minutes) required to consume half of A is :

(A) 5 (B) 10 (C) 1 (D) 100

Q73. JEE Main 2018 (Online) 16th April Morning Slot

If 50% of a reaction occurs in 100 second and 75% of the reaction occurs in 200 second, the order of this reaction is :

Open solution

(A) Zero (B) 1 (C) 2 (D) 3

Q74. JEE Main 2018 (Offline)

At 518^oC the rate of decomposition of a sample of gaseous acetaldehyde initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is

Open solution

(A) 0 (B) 2 (C) 3 (D) 1

Q75. JEE Main 2018 (Online) 15th April Evening Slot

For a first order reaction, A $\rightarrow$ P, $t_{1/2}$ (half-life) is 10 days. The time required for $\frac{1}{4}^{th}$ conversion of A (in days) is : (ln 2 = 0.693, ln 3 = 1.1)

Open solution

(A) 5 (B) 3.2 (C) 4.1 (D) 2.5

Q76. JEE Main 2018 (Online) 15th April Morning Slot

N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg. The pressure of the gaseous mixture after 100 minute at constant temperature will be :

Open solution

(A) 175.0 mmHg (B) 116.25 mmHg (C) 136.25 mmHg (D) 106.25 mmHg

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Q77. JEE Main 2017 (Online) 9th April Morning Slot

The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is :

(Assume activation energy and preexponential factor are independent of temperature; ln 2 = 0.693; R = 8.314 J mol^{-1} K^{-1})

(A) 107.2 kJ mol^{-1} (B) 53.6 kJ mol^{-1} (C) 26.8 kJ mol^{-1} (D) 214.4 kJ mol^{-1}

Open solution

Q78. JEE Main 2017 (Online) 8th April Morning Slot

The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be Increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A.

(A) 9.84 K (B) 4.92 K (C) 2.45 K (D) 19.67 K

Open solution

Q79. JEE Main 2017 (Offline)

Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to :

(R = 8.314 J mol^{-1} K^{-1})

(A) 12 (B) 6 (C) 4 (D) 8

Open solution

Q80. JEE Main 2016 (Online) 10th April Morning Slot

The rate law for the reaction below is given by the expression k [A] [B]

A + B $\rightarrow$ Product

If the concentration of B is increased from 0.1 to 0.3 mole, keeping the value of A at 0.1 mole, the rate constant will be :

(A) k (B) k/3 (C) 3k (D) 9k

Open solution

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Q81. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below :

$O_3(g) + Cl^{\bullet}(g) \rightarrow O_2(g) + ClO^{\bullet}(g)$ . . . . . .(i)

$k_i = 5.2 \times 10^9$ L mol^{-1} s^{-1}

$ClO^{\bullet}(g) + O^{\bullet}(g) \rightarrow O_2(g) + Cl^{\bullet}(g)$ . . . . . (ii)

$k_{ii} = 2.6 \times 10^{10}$ L mol^{-1} s^{-1}

The closest rate constant for the overall reaction $O_3(g) + O^{\bullet}(g) \rightarrow 2\,O_2(g)$ is :

(A) 5.2×10^9 L mol^{-1} s^{-1}

(B) 2.6×10^{10} L mol^{-1} s^{-1}

(C) 3.1×10^{10} L mol^{-1} s^{-1}

(D) 1.4×10^{20} L mol^{-1} s^{-1}

Q82. JEE Main 2016 (Offline)

Open solution

Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :

(A) 6.93×10^{-4} mol min^{-1}

(B) 6.96×10^{-2} mol min^{-1}

(C) 1.34×10^{-2} mol min^{-1}

(D) 2.66 L min^{-1} at STP

Q83. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

For the equilibrium , $A(g) \rightleftharpoons B(g)$, ΔH is $-$ 40 kJ/mol. If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :

(A) E_f = 60 kJ/mol; E_b = 100 kJ/mol

(B) E_f = 30 kJ/mol; E_b = 70 kJ/mol

(C) E_f = 80 kJ/mol; E_b = 120 kJ/mol

(D) E_f = 70 kJ/mol; E_b = 30 kJ/mol

Q84. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

$A + 2B \rightarrow C$, the rate equation for this reaction is given as

Rate = k [A][B].

If the concentration of A is kept the same but that of B is doubled what will happen to the rate itself ?

(A) halved (B) the same (C) doubled (D) quadrupled

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Q85. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

The reaction

$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mm Hg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant) :

(A) 106.25 mm Hg (B) 116.25 mm Hg (C) 125 mm Hg (D) 150 mm Hg

Q86. JEE Main 2015 (Offline)

Open solution

Higher order (>3) reactions are rare due to

(A) increase in entropy and activation energy as more molecules are involved

(B) shifting of equilibrium towards reactants due to elastic collisions

(C) loss of active species on collision

(D) low probability of simultaneous collision of all the reacting species

Q87. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

For the reaction, $3A + 2B \rightarrow C + D$,
the differential rate law can be written as :

(A) $\frac{1}{3}\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

(B) $-\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

(C) $+\frac{1}{3}\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

(D) $-\frac{1}{3}\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

Q88. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

The rate coefficient (k) for a particular reaction is 1.3×10^{-4} M^{-1} s^{-1} at 100°C, and 1.3×10^{-3} M^{-1} s^{-1} at 150°C. What is the energy of activation (E_a) (in kJ) for this reaction ? (R = molar gas constant = 8.314 JK^{-1} mol^{-1})

(A) 16 (B) 60 (C) 99 (D) 132

Q89. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

In the reaction of formation of sulphur trioxide by contact process
$2SO_2 + O_2 \rightleftharpoons 2SO_3$ the rate of reaction was measured as
$\frac{d[O_2]}{dt} = -2.5 \times 10^{-4}$ mol L^{-1} s^{-1}.
The rate of reaction in terms of $[SO_2]$ in mol L^{-1} s^{-1}

(A) -1.25×10^{-4} (B) -2.50×10^{-4} (C) -3.75×10^{-4} (D) -5.00×10^{-4}

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Q90. JEE Main 2014 (Online) 11th April Morning Slot

For the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$, the rate equation can be expressed in two ways $-\frac{d[N_2O_5]}{dt} = k\,[N_2O_5]$ and $+\frac{d[NO_2]}{dt} = k'\,[N_2O_5]$

k and k' are related as :

(A) k = k' (B) 2k = k' (C) k = 2k' (D) k = 4k'

Open solution

Q91. JEE Main 2014 (Online) 9th April Morning Slot

The half-life period of a first order reaction is 15 minutes. The amount of substance left after one hour will be :

(A) $\frac{1}{4}$ of the original amount (B) $\frac{1}{8}$ of the original amount

(C) $\frac{1}{16}$ of the original amount (D) $\frac{1}{32}$ of the original amount

Open solution

Q92. JEE Main 2014 (Offline)

For the non – stoichiometre reaction 2A + B $\rightarrow$ C + D, the following kinetic data were obtained in three separate experiments, all at 298 K.

Initial Concentration (A)	Initial Concentration (B)	Initial rate of formation of C (mol L^{-1} s^{-1})
0.1 M	0.1 M	1.2×10^{-3}
0.1 M	0.2 M	1.2×10^{-3}
0.2 M	0.1 M	2.4×10^{-3}

The rate law for the formation of C is:

(A) $\frac{dc}{dt} = k[A][B]^2$ (B) $\frac{dc}{dt} = k[A]$ (C) $\frac{dc}{dt} = k[A][B]$ (D) $\frac{dc}{dt} = k[A^2][B]$

Open solution

Q93. JEE Main 2013 (Online) 25th April Morning Slot

A radioactive isotope having a half-life period of 3 days was received after 12 days. If 3 g of the isotope is left in the container, what would be the initial mass of the isotope?

(A) 12 g (B) 36 g (C) 48 g (D) 24 g

Open solution

Q94. JEE Main 2013 (Online) 23th April Morning Slot

The rate constant of a zero order reaction is 2.0×10^{-2} mol L^{-1} s^{-1}. If the concentration of the reactant after 25 seconds is 0.5M. What is the initial concentration?

(A) 0.5M (B) 1.25M (C) 12.5M (D) 1.0M

Open solution

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Q95. JEE Main 2013 (Online) 22th April Morning Slot

The reaction X $\rightarrow$ Y is an exothermic reaction. Activation energy of the reaction for X into Y is 150 kJ mol^{-1}. Enthalpy of reaction is 135 kJ mol^{-1}. The activation energy for the reverse reaction, Y $\rightarrow$ X will be :

(A) 280 kJ mol^{-1} (B) 285 kJ mol^{-1} (C) 270 kJ mol^{-1} (D) 15 kJ mol^{-1}

Open solution

Q96. JEE Main 2013 (Online) 9th April Morning Slot

The instantaneous rate of disappearance of MnO_4^- ion in the following reaction is 4.56×10^{-3} Ms^{-1}

$2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

The rate of appearance of I_2 is :

(A) 4.56×10^{-4} Ms^{-1} (B) 1.14×10^{-2} Ms^{-1} (C) 1.14×10^{-3} Ms^{-1} (D) 5.7×10^{-3} Ms^{-1}

Open solution

Q97. JEE Main 2013 (Offline)

The rate of a reaction doubles when its temperature changes from 300K to 310K. Activation energy of such a reaction will be:(R = 8.314 JK^{-1} mol^{-1} and log 2 = 0.301)

(A) 48.6 kJ mol^{-1} (B) 58.5 kJ mol^{-1} (C) 60.5 kJ mol^{-1} (D) 53.6 kJ mol^{-1}

Open solution

Q98. AIEEE 2012 (Online) 26th May

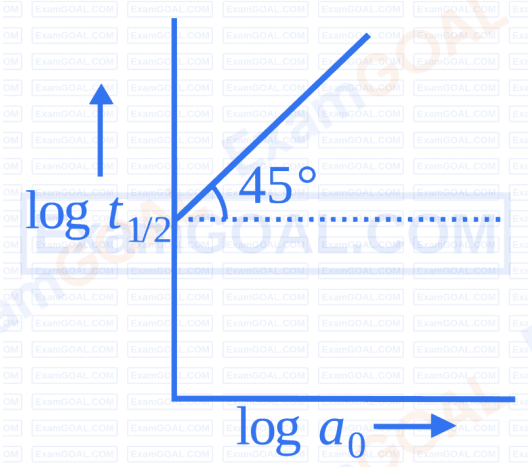
The activation energy for a reaction which doubles the rate when the temperature is raised from 298 K to 308 K is

(A) 59.2 kJ mol^{-1} (B) 39.2 kJ mol^{-1} (C) 52.9 kJ mol^{-1} (D) 29.5 kJ mol^{-1}

Open solution

Q99. AIEEE 2012 (Online) 19th May

For a reaction $A \rightarrow$ Products, a plot of $\log t_{1/2}$ versus $\log a_0$ is shown in the figure.



Open solution

If the initial concentration of A is represented by a_0, the order of the reaction is

(A) one (B) zero (C) two (D) three

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Q100. AIEEE 2012 (Online) 12th May

In a chemical reaction A is converted into B. The rates of reaction, starting with initial concentrations of A as 2×10^{-3} M and 1×10^{-3} M, are equal to $2.40 \times 10^{-4}\,\mathrm{M\,s^{-1}}$ and $0.60 \times 10^{-4}\,\mathrm{M\,s^{-1}}$ respectively. The order of reaction with respect to reactant A will be

(A) 0 (B) 1.5 (C) 1 (D) 2

Open solution

Q101. AIEEE 2012 (Online) 7th May

Reaction rate between two substance A and B is expressed as following:
rate $= k[A]^n[B]^m$
If the concentration of A is doubled and concentration of B is made half of initial concentration, the ratio of the new rate to the earlier rate will be:

(A) $m+n$ (B) $n-m$ (C) $\frac{1}{2^{(m+n)}}$ (D) $2^{(n-m)}$

Open solution

Q102. AIEEE 2012

For a first order reaction, (A) $\rightarrow$ products, the concentration of A changes from 0.1 M to 0.025 M in 40 minutes. The rate of reaction when the concentration of A is 0.01 M is :

(A) 1.73 x 10^{-5} M/ min (B) 3.47 x 10^{-4} M/min

(C) 3.47 x 10^{-5} M/min (D) 1.73 x 10^{-4} M/min

Open solution

Q103. AIEEE 2011

The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C , the rate of the reaction increases by about :

(A) 24 times (B) 32 times (C) 64 times (D) 10 times

Open solution

Q104. AIEEE 2010

The time for half life period of a certain reaction A $\rightarrow$ products is 1 hour. When the initial concentration of the reactant 'A', is 2.0 mol L^{-1}, how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction ?

(A) 4 h (B) 0.5 h (C) 0.25 h (D) 1 h

Open solution

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Q105. AIEEE 2010

Open solution

Consider the reaction :

$Cl_2(aq) + H_2S(aq) \rightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$

The rate equation for this reaction is rate = k $[Cl_2]$ $[H_2S]$

Which of these mechanisms is/are consistent with this rate equation?

(A) $Cl_2 + H_2S \rightarrow H^+ + Cl^- + Cl^+ + HS^-$ (slow)

$Cl^+ + HS^- \rightarrow H^+ + Cl^- + S$ (fast)

(B) $H_2S \Leftrightarrow H^+ + HS^-$ (fast equilibrium)

$Cl_2 + HS^- \rightarrow 2Cl^- + H^+ + S$ (slow)

(A) B only **(B)** Both A and B **(C)** Neither A nor B **(D)** A only

Q106. AIEEE 2009

Open solution

The half life period of a first order chemical reaction is 6.93 minutes. The time required for the completion of 99% of the chemical reaction will be (log 2=0.301) :

(A) 230.3 minutes **(B)** 23.03 minutes **(C)** 46.06 minutes **(D)** 460.6 minutes

Q107. AIEEE 2008

Open solution

For a reaction $\frac{1}{2}A \rightarrow 2B$ rate of disappearance of 'A' is related to the rate of appearance of 'B' by the expression

(A) $-\frac{d[A]}{dt} = \frac{1}{2}\frac{d[B]}{dt}$ **(B)** $-\frac{d[A]}{dt} = \frac{1}{4}\frac{d[B]}{dt}$

(C) $-\frac{d[A]}{dt} = \frac{d[B]}{dt}$ **(D)** $-\frac{d[A]}{dt} = 4\frac{d[B]}{dt}$

Q108. AIEEE 2007

Open solution

The energies of activation for forward and reverse reactions for $A_2 + B_2 \leftrightharpoons 2AB$ are 180 kJ mol^{-1} and 200 kJ mol^{-1} respectively. The presence of catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol^{-1}. The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of catalyst will be (in kJ mol^{-1})

(A) 300 **(B)** 120 **(C)** 200 **(D)** 20

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Q109. AIEEE 2007

Consider the reaction, 2A + B $\rightarrow$ products. When concentration of B alone was doubled, the half-life did not change. When the concentration of A alone was doubled, the rate increased by two times. The unit of rate constant for this reaction is

(A) L mol^{-1} s^{-1} (B) no unit (C) mol L^{-1} s^{-1} (D) s^{-1}

Open solution

Q110. AIEEE 2007

A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial activity is ten times the permissible value, after how many days will it be safe to enter the room?

(A) 1000 days (B) 300 days (C) 10 days (D) 100 days

Open solution

Q111. AIEEE 2007

Which of the following nuclear reactions will generate an isotope?

(A) neutron particle emission (B) positron emission
(C) α-particle emission (D) β-particle emission

Open solution

Q112. AIEEE 2006

A reaction was found to be second order with respect to the concentration of carbon monoxide. If the concentration of carbon monoxide is doubled, with everything else kept the same, the rate of reaction will

(A) remain unchanged (B) triple
(C) increase by a factor of 4 (D) double

Open solution

Q113. AIEEE 2006

Rate of a reaction can be expressed by Arrhenius equation as:

$$k = A\,e^{-E/RT}$$

In this equation, E represents

(A) the energy above which all the colliding molecules will react
(B) the energy below which colliding molecules will not react
(C) the total energy of the reacting molecules at a temperature, T
(D) the fraction of molecules with energy greater than the activation energy of the reaction

Open solution

Q114. AIEEE 2006

In the transformation of $^{238}_{92}U$ to $^{234}_{92}U$, if one emission is an α-particle, what should be the other emission(s)?

(A) Two β^- (B) Two β^- and one β^+
(C) One β^- and one γ (D) One β^+ and One β^-

Open solution

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Q115. AIEEE 2006

The following mechanism has been proposed for the reaction of NO with Br_2 to form NOBr:

$NO(g) + Br_2(g) \rightleftharpoons NOBr_2(g)$

$NOBr_2(g) + NO(g) \rightarrow 2NOBr(g)$

If the second step is the rate determining step, the order of the reaction with respect to NO(g) is

(A) 1 (B) 0 (C) 3 (D) 2

Open solution

Q116. AIEEE 2005

Consider an endothermic reaction, $X \rightarrow Y$ with the activation energies E_b and E_f for the backward and forward reactions, respectively. In general :

(A) $E_b < E_f$

(B) $E_b > E_f$

(C) $E_b = E_f$

(D) There is no definite relation between E_b and E_f

Open solution

Q117. AIEEE 2005

Hydrogen bomb is based on the principle of

(A) Nuclear fission (B) Natural radioactivity

(C) Nuclear fusion (D) Artificial radioactivity

Open solution

Q118. AIEEE 2005

A reaction involving two different reactants can never be

(A) Unimolecular reaction (B) First order reaction

(C) second order reaction (D) Bimolecular reaction

Open solution

Q119. AIEEE 2005

The photon of hard gamma radiation knocks a proton out of $^{24}_{12}Mg$ nucleus to form

(A) the isotope of parent nucleus (B) the isobar of parent nucleus

(C) the nuclide $^{23}_{11}Na$ (D) the isobar of $^{23}_{11}Na$

Open solution

Q120. AIEEE 2005

$t_{1/4}$ can be taken as the time taken for the concentration of a reactant to drop to $\frac{3}{4}$ of its initial value. If the rate constant for a first order reaction is K, the $t_{1/4}$ can be written as

(A) 0.10 / K (B) 0.29 / K (C) 0.69 / K (D) 0.75 / K

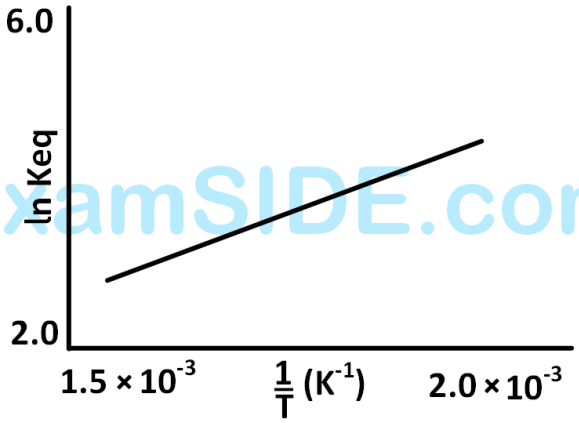
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Q121. AIEEE 2005

A schematic plot of $ln\ K_{eq}$ versus inverse of temperature for a reaction is shown below

Open solution



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The reaction must be

(A) highly spontaneous at ordinary temperature

(B) one with negligible enthalpy change

(C) endothermic

(D) exothermic

Q122. AIEEE 2004

In a first order reaction, the concentration of the reactant decreases from 0.8 M to 0.4 M in 15 minutes. The time taken for the concentration to change from 0.1 M to 0.025 M is

(A) 30 minutes (B) 60 minutes (C) 7.5 minutes (D) 15 minutes

Open solution

Q123. AIEEE 2004

The rate equation for the reaction 2A + B $\rightarrow$ C is found to be: rate = k[A][B]. The correct statement in relation to this reaction is that the

(A) unit of k must be s^{-1}

(B) value of k is independent of the initial concentration of A and B

(C) rate of formation of C is twice the rate of disappearance of A

(D) $t_{1/2}$ is a constant

Open solution

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Q124. AIEEE 2004

Consider the following nuclear reactions
$^{238}_{92}M \rightarrow {}^{X}_{Y}N + 2^{4}_{2}He$
$^{X}_{Y}N \rightarrow {}^{A}_{B}L + 2\beta^{+}$
The number of neutrons in the element L is

(A) 140 (B) 144 (C) 142 (D) 146

Open solution

Q125. AIEEE 2004

The half – life of a radioisotope is four hours. If the initial mass of the isotope was 200 g, the mass remaining after 24 hours undecayed is

(A) 1.042 g (B) 4.167 g (C) 3.125 g (D) 2.084 g

Open solution

Q126. AIEEE 2003

The radionucleide $^{234}_{90}Th$ undergoes two successive β-decays followed by one α-decay. The atomic number and the mass number respectively of the resulting radionucleide are

(A) 94 and 230 (B) 90 and 230 (C) 92 and 230 (D) 92 and 234

Open solution

Q127. AIEEE 2003

The half-life of a radioactive isotope is three hours. If the initial mass of the isotope were 256 g, the mass of it remaining undecayed after 18 hours would be

(A) 8.0 g (B) 12.0 g (C) 16.0 g (D) 4.0 g

Open solution

Q128. AIEEE 2003

In respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statements is correct?

(A) A is adsorption factor (B) E_a is energy of activation

(C) R is Rydberg's constant (D) k is equilibrium constant

Open solution

Q129. AIEEE 2003

The rate law for a reaction between the substances A and B is given by Rate = $k[A]^n [B]^m$ On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as

(A) (m + n) (B) (n - m) (C) $2^{(n-m)}$ (D) $\frac{1}{2^{(m+n)}}$

Open solution

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Q130. AIEEE 2003

For the reaction system:

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ volume is suddenly reduce to half its value by increasing the pressure on it. If the reaction is of first order with respect to O_2 and second order with respect to NO, the rate of reaction will

(A) diminish to one-eighth of its initial value
(B) increase to eight times of its initial value
(C) increase to four times of its initial value
(D) diminish to one-fourth of its initial value

Open solution

Q131. AIEEE 2002

Units of rate constant of first and zero order reactions in terms of molarity M unit are respectively

(A) sec^{-1}, $Msec^{-1}$ (B) sec^{-1}, M (C) $Msec^{-1}$, sec^{-1} (D) M, sec^{-1}

Open solution

Q132. AIEEE 2002

For the reaction $A + 2B \rightarrow C$, rate is given by $R = [A][B]^2$ then the order of the reaction is

(A) 3 (B) 6 (C) 5 (D) 7

Open solution

Q133. AIEEE 2002

The differential rate law for the reaction $H_2 + I_2 \rightarrow 2HI$ is

(A) $-\frac{d[H_2]}{dt} = -\frac{d[I_2]}{dt} = -\frac{d[HI]}{dt}$
(B) $\frac{d[H_2]}{dt} = \frac{d[I_2]}{dt} = \frac{d[HI]}{dt}$
(C) $\frac{1}{2}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[I_2]}{dt} = -\frac{d[HI]}{dt}$
(D) $-2\frac{d[H_2]}{dt} = -2\frac{d[I_2]}{dt} = \frac{d[HI]}{dt}$

Open solution

Q134. AIEEE 2002

If half-life of a substance is 5 yrs, then the total amount of substance left after 15 years, when initial amount is 64 grams is

(A) 16 grams (B) 2 grams (C) 32 grams (D) 8 grams

Open solution

Q135. AIEEE 2002

The integrated rate equation is $Rt = \log C_0 - \log C_t$. The straight line graph is obtained by plotting

(A) time vs $\log C_t$ (B) $\frac{1}{time}$ vs C_t
(C) time vs C_t (D) $\frac{1}{time}$ vs $\frac{1}{C_t}$

Open solution

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Q136. AIEEE 2002

β - particle is emitted in radioactivity by

(A) conversion of proton to neutron

(B) from outermost orbit

(C) conversion of neutron to proton

(D) β -particle is not emitted

Open solution

Numerical 105 questions

Q137. JEE Main 2026 (Online) 6th April Evening Shift

Decomposition of a hydrocarbon follows the equation $\mathrm{k} = \left(5.5 \times 10^{11}\ \mathrm{s}^{-1}\right) \mathrm{e}^{\frac{-28000\ \mathrm{K}}{\mathrm{T}}}$. The activation energy of reaction is ____ kJmol^{-1}. (Nearest Integer)

Given : $\mathrm{R} = 8.3\ \mathrm{J\ K}^{-1}\ \mathrm{mol}^{-1}$

Open solution

Q138. JEE Main 2026 (Online) 6th April Morning Shift

Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is

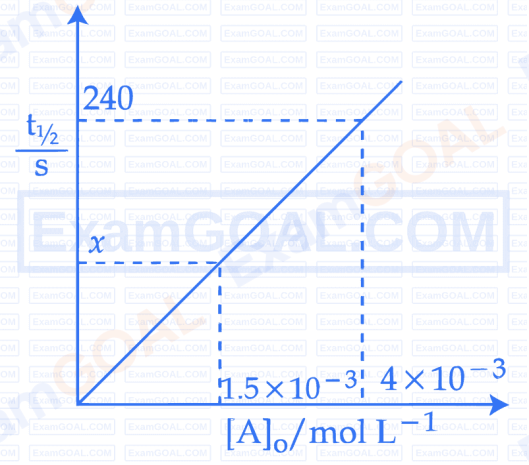
____ . (Nearest integer)

(Given : $\log 2 = 0.3010$ and $\log 3 = 0.4771$)

Open solution

Q139. JEE Main 2026 (Online) 5th April Evening Shift

For a reaction $\mathrm{A} \rightarrow \mathrm{P}$ at T K , the half life $\left(t_{1/2}\right)$ is plotted as a function of initial concentration $[\mathrm{A}]_0$ of A as given below.



The value of x in the given figure is ____ s (Nearest integer)

Open solution

Q140. JEE Main 2026 (Online) 5th April Morning Shift

If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be ____ minutes.

(Given : $\log 2 = 0.3010$)

Open solution

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Q141. JEE Main 2026 (Online) 4th April Evening Shift

For a first order reaction $\mathrm{A} \rightarrow \mathrm{B}$

$\mathrm{t/\,min}$	$\mathrm{[A]/M}$
0	0.6500
x	0.0650
20	0.00065

$x =$ ____ min. (Nearest integer)

Open solution

Q142. JEE Main 2026 (Online) 2nd April Evening Shift

Consider the following gas phase reaction being carried out in a closed vessel at $25^{\circ}\mathrm{C}$.

$2A(g) \longrightarrow 4B(g) + C(g)$

time (min)	total pressure of the system (mm Hg)
30	300
∞	600

The pressure of C(g) at 30 minutes time interval would be _____ mm Hg. (nearest integer)

Open solution

Q143. JEE Main 2026 (Online) 2nd April Morning Shift

For reaction A → P, rate constant $k = 1.5 \times 10^3\ \mathrm{s}^{-1}$ at $27^{\circ}\mathrm{C}$

If activation energy for the above reaction is $60\ \mathrm{kJ\ mol}^{-1}$, then the temperature (in $^{\circ}\mathrm{C}$) at which rate constant, $k = 4.5 \times 10^3\ \mathrm{s}^{-1}$ is _____. (Nearest integer)

Given : $\log 2 = 0.30$, $\log 3 = 0.48$, $R = 8.3\ \mathrm{J\ K}^{-1}\ \mathrm{mol}^{-1}$, $\ln 10 = 2.3$

Open solution

Q144. JEE Main 2026 (Online) 28th January Evening Shift

A → B (first reaction)

C → D (second reaction)

Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300 K. At 500 K, 50% of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the same temperature; then rate constant for the second reaction at 300 K is _____ × 10^{-1} hour^{-1} (nearest integer).

Open solution

Q145. JEE Main 2026 (Online) 24th January Evening Shift

The half-life of $^{65}\mathrm{Zn}$ is 245 days. After x days, 75% of original activity remained. The value of x in days is ____ . (Nearest integer)

(Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

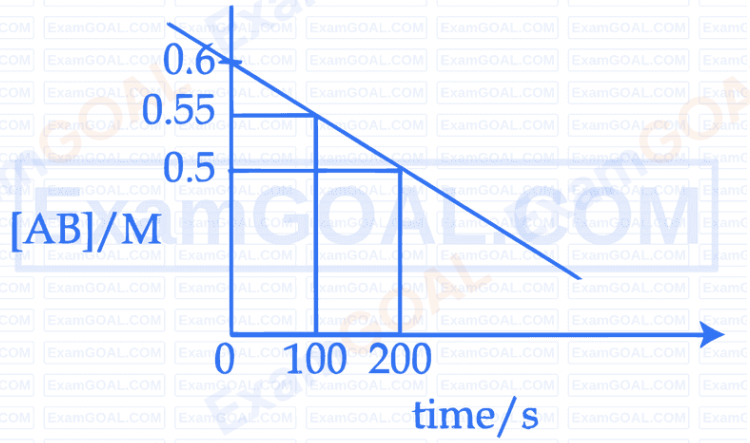
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Q146. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

For the thermal decomposition of reactant $AB(g)$, the following plot is constructed.



The half life of the reaction is ' x ' min.

$x =$ ____ min. (Nearest integer)

Q147. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Consider $A \xrightarrow{k_1} B$ and $C \xrightarrow{k_2} D$ are two reactions. If the rate constant (k_1) of the $A \longrightarrow B$ reaction can be expressed by the following equation $\log_{10} k = 14.34 - \frac{1.5\times10^4}{T/K}$ and activation energy of $C \longrightarrow D$ reaction (Ea_2) is $\frac{1}{5}$ th of the $A \longrightarrow B$ reaction (Ea_1), then the value of (Ea_2) is ____ $kJmol^{-1}$. (Nearest Integer)

Q148. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

The temperature at which the rate constants of the given below two gaseous reactions become equal is ____ K. (Nearest integer)

$X \longrightarrow Y, \quad k_1 = 10^6 e^{\frac{-30000}{T}}$

$P \longrightarrow Q, \quad k_2 = 10^4 e^{\frac{-24000}{T}}$

Given : $\ln 10 = 2.303$

Q149. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by $20\ kJ\ mol^{-1}$. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K , then $\ln \frac{k_2}{k_1}$ will be ____ . (nearest integer) $\left[R = 8.3\ J\ K^{-1}\ mol^{-1}\right]$

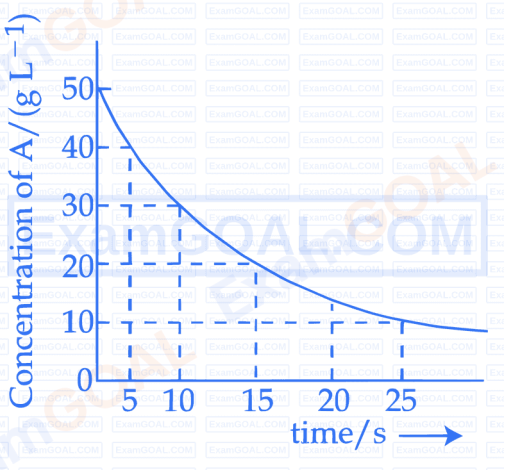
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Q150. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

For the reaction $A \rightarrow B$ the following graph was obtained. The time required (in seconds) for the concentration of A to reduce to $2.5\ g\ L^{-1}$ (if the initial concentration of A was $50\ g\ L^{-1}$) is . (Nearest integer)

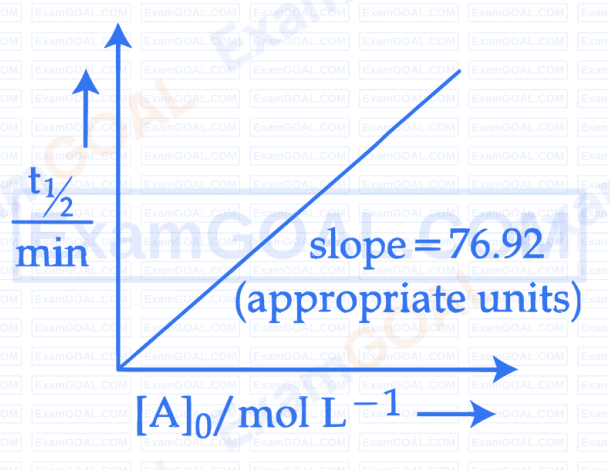
Given : $\log 2 = 0.3010$



Q151. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

For the reaction A $\rightarrow$ products.



The concentration of A at 10 minutes is _____ $\times 10^{-3}\ mol\ L^{-1}$ (nearest integer). The reaction was started with $2.5\ mol\ L^{-1}$ of A .

Q152. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Consider a complex reaction taking place in three steps with rate constants k_1, k_2 and k_3 respectively. The overall rate constant k is given by the expression $k = \sqrt{\frac{k_1 k_3}{k_2}}$. If the activation energies of the three steps are 60, 30 and $10\ kJ\ mol^{-1}$ respectively, then the overall energy of activation in $kJmol^{-1}$ is _____ . (Nearest integer)

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Q153. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

For the thermal decomposition of $N_2O_5(g)$ at constant volume, the following table can be formed, for the reaction mentioned below.

$2\,N_2O_5(g) \rightarrow 2\,N_2O_4(g) + O_2(g)$

Sr. No.	Time/s	Total pressure/(atm)
1	0	0.6
2	100	'x'

$x =$ _____ $\times 10^{-3}$ atm [nearest integer]

Given : Rate constant for the reaction is $4.606 \times 10^{-2}\,s^{-1}$.

Q154. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

$A \rightarrow B$

The molecule A changes into its isomeric form B by following a first order kinetics at a temperature of 1000 K . If the energy barrier with respect to reactant energy for such isomeric transformation is $191.48\,kJ\,mol^{-1}$ and the frequency factor is 10^{20}, the time required for 50% molecules of A to become B is _____ picoseconds (nearest integer). $\left[R = 8.314\,J\,K^{-1}\,mol^{-1}\right]$

Q155. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Consider the following first order gas phase reaction at constant temperature $A(g) \rightarrow 2B(g) + C(g)$

If the total pressure of the gases is found to be 200 torr after 23 sec. and 300 torr upon the complete decomposition of A after a very long time, then the rate constant of the given reaction is _____ $\times 10^{-2}\,s^{-1}$ (nearest integer)

[Given : $\log_{10}(2) = 0.301$]

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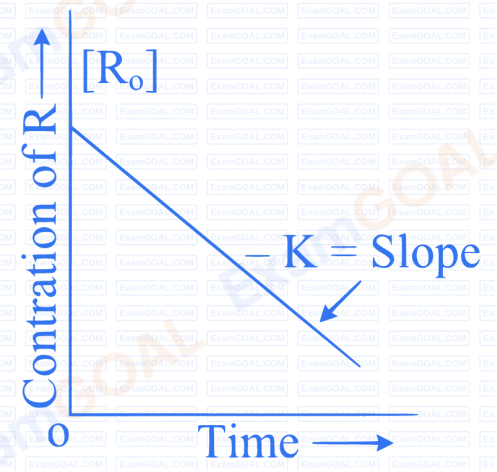
Q156. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Given below are two statements :

Statement I : The rate law for the reaction $A + B \rightarrow C$ is rate $(r) = k[A]^2[B]$. When the concentration of both A and B is doubled, the reaction rate is increased "x" times.

Statement II :



The figure is showing "the variation in concentration against time plot" for a "y" order reaction.

The value of $x + y$ is _____.

Q157. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Consider the following reaction

$A + B \rightarrow C$

The time taken for A to become $1/4^{th}$ of its initial concentration is twice the time taken to become $1/2$ of the same. Also, when the change of concentration of B is plotted against time, the resulting graph gives a straight line with a negative slope and a positive intercept on the concentration axis.

The overall order of the reaction is _____.

Q158. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Consider the two different first order reactions given below

$A + B \rightarrow C$ (Reaction 1)

$P \rightarrow Q$ (Reaction 2)

The ratio of the half life of Reaction 1 : Reaction 2 is $5 : 2$ If t_1 and t_2 represent the time taken to complete $2/3^{rd}$ and $4/5^{th}$ of Reaction 1 and Reaction 2 , respectively, then the value of the ratio $t_1 : t_2$ is _____ $\times 10^{-1}$ (nearest integer). [Given : $\log_{10}(3) = 0.477$ and $\log_{10}(5) = 0.699$]

Q159. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Time required for 99.9% completion of a first order reaction is _____ times the time required for completion of 90% reaction.(nearest integer)

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Q160. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

Consider the following single step reaction in gas phase at constant temperature.

$2\,A_{(g)} + B_{(g)} \rightarrow C_{(g)}$

The initial rate of the reaction is recorded as r_1 when the reaction starts with $1.5\ atm$ pressure of A and $0.7\ atm$ pressure of B. After some time, the rate r_2 is recorded when the pressure of C becomes $0.5\ atm$. The ratio $r_1 : r_2$ is _____ $\times 10^{-1}$. (Nearest integer)

Q161. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

During Kinetic study of reaction $2A + B \rightarrow C + D$, the following results were obtained :

	A[M]	B[M]	initial rate of formation of D
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Based on above data, overall order of the reaction is _____.

Q162. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Consider the following reaction, the rate expression of which is given below

$A + B \rightarrow C$

rate $= k[A]^{1/2}[B]^{1/2}$

The reaction is initiated by taking $1\ M$ concentration of A and B each. If the rate constant (k) is $4.6 \times 10^{-2}\ s^{-1}$, then the time taken for A to become $0.1\ M$ is _____ sec. (nearest integer)

Q163. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Consider the following transformation involving first order elementary reaction in each step at constant temperature as shown below.

$$A + B \underset{\text{Step 3}}{\overset{\text{Step 1}}{\rightleftharpoons}} C \xrightarrow{\text{Step 2}} P$$

Some details of the above reactions are listed below.

Step	Rate constant (sec^{-1})	Activation energy (kJ mol^{-1})
1	k_1	300
2	k_2	200
3	k_3	Ea_3

If the overall rate constant of the above transformation (k) is given as $k = \frac{k_1 k_2}{k_3}$ and the overall activation energy (E_a) is $400\ kJ\ mol^{-1}$, then the value of Ea_3 is _____ $kJ\ mol^{-1}$ (nearest integer)

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Q164. JEE Main 2024 (Online) 1st February Evening Shift

The following data were obtained during the first order thermal decomposition of a gas A at constant volume :

$$A(g) \rightarrow 2\,B(g) + C(g)$$

S.No.	Time /s	Total pressure /(atm)
1.	0	0.1
2.	115	0.28

The rate constant of the reaction is _____ $\times 10^{-2}\,s^{-1}$ (nearest integer)

Open solution

Q165. JEE Main 2024 (Online) 1st February Morning Shift

The ratio of $\frac{^{14}C}{^{12}C}$ in a piece of wood is $\frac{1}{8}$ part that of atmosphere. If half life of ^{14}C is 5730 years, the age of wood sample is _____ years.

Open solution

Q166. JEE Main 2024 (Online) 31st January Evening Shift

$r = k[A]$ for a reaction, 50% of A is decomposed in 120 minutes. The time taken for 90% decomposition of A is _____ minutes.

Open solution

Q167. JEE Main 2024 (Online) 30th January Evening Shift

NO_2 required for a reaction is produced by decomposition of N_2O_5 in CCl_4 as by equation

$$2\,N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$$

The initial concentration of N_2O_5 is $3\ mol\ L^{-1}$ and it is $2.75\ mol\ L^{-1}$ after 30 minutes.

The rate of formation of NO_2 is $x \times 10^{-3}\ mol\ L^{-1}\ min^{-1}$, value of x is _____. (nearest integer)

Open solution

Q168. JEE Main 2024 (Online) 30th January Morning Shift

The rate of a first order reaction is $0.04\ mol\ L^{-1}\ s^{-1}$ at 10 minutes and $0.03\ mol\ L^{-1}\ s^{-1}$ at 20 minutes after initiation. Half life of the reaction is _____ minutes.

(Given $\log 2 = 0.3010, \log 3 = 0.4771$)

Open solution

Q169. JEE Main 2024 (Online) 29th January Evening Shift

The half-life of radioisotope bromine - 82 is 36 hours. The fraction which remains after one day is _____ $\times 10^{-2}$.

(Given antilog $0.2006 = 1.587$)

Open solution

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Q170. JEE Main 2024 (Online) 29th January Morning Shift

For a reaction taking place in three steps at same temperature, overall rate constant $K = \frac{K_1 K_2}{K_3}$. If Ea_1, Ea_2 and Ea_3 are 40, 50 and $60\ kJ/mol$ respectively, the overall Ea is _____ kJ/mol.

Open solution

Q171. JEE Main 2024 (Online) 27th January Evening Shift

Time required for completion of 99.9% of a First order reaction is _____ times of half life $(t_{1/2})$ of the reaction.

Open solution

Q172. JEE Main 2024 (Online) 27th January Morning Shift

Consider the following data for the given reaction

$2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$

	1	2	3
$HI\ (mol\ L^{-1})$	0.005	0.01	0.02
$Rate\ (mol\ L^{-1}s^{-1})$	7.5×10^{-4}	3.0×10^{-3}	1.2×10^{-2}

The order of the reaction is _____.

Open solution

Q173. JEE Main 2023 (Online) 15th April Morning Shift

For a reversible reaction $A \rightleftharpoons B$, the $\Delta H_{\text{forward reaction}} = 20\ kJ\ mol^{-1}$. The activation energy of the uncatalysed forward reaction is $300\ kJ\ mol^{-1}$. When the reaction is catalysed keeping the reactant concentration same, the rate of the catalysed forward reaction at $27^{\circ}C$ is found to be same as that of the uncatalysed reaction at $327^{\circ}C$.

The activation energy of the catalysed backward reaction is _____ $kJ\ mol^{-1}$.

Open solution

Q174. JEE Main 2023 (Online) 13th April Evening Shift

A(g) $\rightarrow$ 2B(g) + C(g) is a first order reaction. The initial pressure of the system was found to be 800 mm Hg which increased to 1600 mm Hg after 10 min. The total pressure of the system after 30 min will be _____ mm Hg. (Nearest integer)

Open solution

Q175. JEE Main 2023 (Online) 13th April Morning Shift

$t_{87.5}$ is the time required for the reaction to undergo 87.5% completion and t_{50} is the time required for the reaction to undergo 50% completion. The relation between $t_{87.5}$ and t_{50} for a first order reaction is $t_{87.5} = x \times t_{50}$ The value of x is _____. (Nearest integer)

Open solution

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Q176. JEE Main 2023 (Online) 12th April Morning Shift

The reaction $2NO + Br_2 \rightarrow 2NOBr$

takes places through the mechanism given below:

$NO + Br_2 \Leftrightarrow NOBr_2$ (fast)

$NOBr_2 + NO \rightarrow 2NOBr$ (slow)

The overall order of the reaction is _____.

Open solution

Q177. JEE Main 2023 (Online) 11th April Morning Shift

$KClO_3 + 6FeSO_4 + 3H_2SO_4 \rightarrow KCl + 3Fe_2(SO_4)_3 + 3H_2O$

The above reaction was studied at $300\ K$ by monitoring the concentration of $FeSO_4$ in which initial concentration was $10M$ and after half an hour became 8.8 M. The rate of production of $Fe_2(SO_4)_3$ is _____ $\times 10^{-6}\ mol\ L^{-1}\ s^{-1}$ (Nearest integer)

Open solution

Q178. JEE Main 2023 (Online) 10th April Evening Shift

The number of **incorrect** statement/s from the following is _____

A. The successive half lives of zero order reactions decreases with time.

B. A substance appearing as reactant in the chemical equation may not affect the rate of reaction

C. Order and molecularity of a chemical reaction can be a fractional number

D. The rate constant units of zero and second order reaction are $mol\ L^{-1}\ s^{-1}$ and $mol^{-1}\ L\ s^{-1}$ respectively

Open solution

Q179. JEE Main 2023 (Online) 10th April Morning Shift

A molecule undergoes two independent first order reactions whose respective half lives are 12 min and 3 min. If both the reactions are occurring then the time taken for the 50% consumption of the reactant is _____ min. (Nearest integer)

Open solution

Q180. JEE Main 2023 (Online) 8th April Morning Shift

The number of given statement/s which is/are correct is _____.

(A) The stronger the temperature dependence of the rate constant, the higher is the activation energy.

(B) If a reaction has zero activation energy, its rate is independent of temperature.

(C) The stronger the temperature dependence of the rate constant, the smaller is the activation energy.

(D) If there is no correlation between the temperature and the rate constant then it means that the reaction has negative activation energy.

Open solution

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Q181. JEE Main 2023 (Online) 1st February Evening Shift

$A \rightarrow B$

The above reaction is of zero order. Half life of this reaction is $50\ \text{min}$. The time taken for the concentration of A to reduce to one-fourth of its initial value is _____ min. (Nearest integer)

Open solution

Q182. JEE Main 2023 (Online) 1st February Morning Shift

A and B are two substances undergoing radioactive decay in a container. The half life of A is 15 min and that of B is 5 min. If the initial concentration of B is 4 times that of A and they both start decaying at the same time, how much time will it take for the concentration of both of them to be same? _____ min.

Open solution

Q183. JEE Main 2023 (Online) 31st January Evening Shift

The rate constant for a first order reaction is $20\ \text{min}^{-1}$. The time required for the initial concentration of the reactant to reduce to its $\frac{1}{32}$ level is _____ $\times 10^{-2}\ \text{min}$. (Nearest integer)

(Given : $\ln 10 = 2.303$ and $\log 2 = 0.3010$)

Open solution

Q184. JEE Main 2023 (Online) 31st January Morning Shift

A $\rightarrow$ B

The rate constants of the above reaction at 200 K and 300 K are 0.03 min^{-1} and 0.05 min^{-1} respectively. The activation energy for the reaction is _____ J (Nearest integer)

(Given : $\ln 10 = 2.3$

$R = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$

$\log 5 = 0.70$

$\log 3 = 0.48$

$\log 2 = 0.30$)

Open solution

Q185. JEE Main 2023 (Online) 30th January Evening Shift

An organic compound undergoes first-order decomposition. If the time taken for the 60% decomposition is $540\ \text{s}$, then the time required for 90% decomposition will be _____ s. (Nearest integer).

Given: $\ln 10 = 2.3; \log 2 = 0.3$

Open solution

Q186. JEE Main 2023 (Online) 30th January Morning Shift

If compound A reacts with B following first order kinetics with rate constant $2.011 \times 10^{-3}\ \text{s}^{-1}$. The time taken by A (in seconds) to reduce from $7\ \text{g}$ to $2\ \text{g}$ will be _____. (Nearest Integer)

$[\log 5 = 0.698, \log 7 = 0.845, \log 2 = 0.301]$

Open solution

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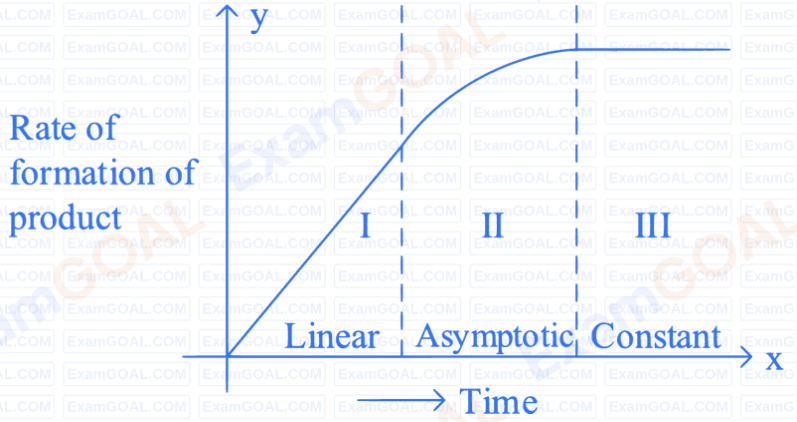
Q187. JEE Main 2023 (Online) 29th January Evening Shift

For conversion of compound A $\rightarrow$ B, the rate constant of the reaction was found to be $4.6 \times 10^{-5}\ \mathrm{L\ mol^{-1}\ s^{-1}}$. The order of the reaction is _____.

Open solution

Q188. JEE Main 2023 (Online) 29th January Morning Shift

For certain chemical reaction $X \rightarrow Y$, the rate of formation of product is plotted against the time as shown in the figure. The number of correct statement/s from the following is _____.



Open solution

(A) Over all order of this reaction is one.

(B) Order of this reaction can't be determined.

(C) In region I and III, the reaction is of first and zero order respectively.

(D) In region-II, the reaction is of first order.

(E) In region-II, the order of reaction is in the range of 0.1 to 0.9.

Q189. JEE Main 2023 (Online) 25th January Evening Shift

A first order reaction has the rate constant, $\mathrm{k} = 4.6 \times 10^{-3}\ \mathrm{s^{-1}}$. The number of correct statement/s from the following is/are _____

Given : $\log 3 = 0.48$

A. Reaction completes in 1000 s.

B. The reaction has a half-life of 500 s.

C. The time required for 10% completion is 25 times the time required for 90% completion.

D. The degree of dissociation is equal to $(1 - \mathrm{e}^{-\mathrm{kt}})$

E. The rate and the rate constant have the same unit.

Open solution

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Q190. JEE Main 2023 (Online) 25th January Morning Shift

For the first order reaction A $\rightarrow$ B, the half life is 30 min. The time taken for 75% completion of the reaction is _____ min. (Nearest integer)

Given : log 2 = 0.3010

log 3 = 0.4771

log 5 = 0.6989

Open solution

Q191. JEE Main 2023 (Online) 24th January Morning Shift

The number of correct statement/s from the following is _____

A. Larger the activation energy, smaller is the value of the rate constant.

B. The higher is the activation energy, higher is the value of the temperature coefficient.

C. At lower temperatures, increase in temperature causes more change in the value of k than at higher temperature

D. A plot of $\ln k$ vs $\frac{1}{T}$ is a straight line with slope equal to $-\frac{E_a}{R}$

Open solution

Q192. JEE Main 2022 (Online) 30th July Evening Shift

$X \rightarrow Y + Z$

X decomposes at 700 K to give Y and Z. The results of two measurements are :

Initial concentration of X / (mol/L)	6.0	12.0
Half life / s	1.0	2.0

The rate constant for the reaction is _____ (in appropriate unit)

Open solution

Q193. JEE Main 2022 (Online) 29th July Evening Shift

Assuming $1\ \mu g$ of trace radioactive element X with a half life of 30 years is absorbed by a growing tree. The amount of X remaining in the tree after 100 years is _____ $\times 10^{-1} \mu g$.

[Given : ln 10 = 2.303; log 2 = 0.30]

Open solution

Q194. JEE Main 2022 (Online) 29th July Morning Shift

The reaction between X and Y is first order with respect to X and zero order with respect to Y.

Experiment	$\frac{[X]}{mol\ L^{-1}}$	$\frac{[Y]}{mol\ L^{-1}}$	$\frac{Initial\ rate}{mol\ L^{-1}\ min^{-1}}$
I	0.1	0.1	2×10^{-3}
II	L	0.2	4×10^{-3}
III	0.4	0.4	$M \times 10^{-3}$
IV	0.1	0.2	2×10^{-3}

Examine the data of table and calculate ratio of numerical values of M and L. (Nearest Integer)

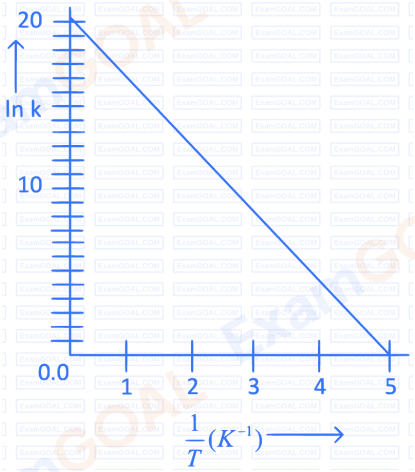
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Q195. JEE Main 2022 (Online) 28th July Evening Shift

For a reaction, given below is the graph of $\ln k$ vs $\frac{1}{T}$. The activation energy for the reaction is equal to _____ cal mol^{-1}. (nearest integer)

(Given : $\text{R} = 2\,\text{cal K}^{-1}\ \text{mol}^{-1}$)



Open solution

Q196. JEE Main 2022 (Online) 28th July Morning Shift

For the given first order reaction

$\text{A} \rightarrow \text{B}$

the half life of the reaction is $0.3010\ \text{min}$. The ratio of the initial concentration of reactant to the concentration of reactant at time $2.0\ \text{min}$ will be equal to _____. (Nearest integer)

Open solution

Q197. JEE Main 2022 (Online) 27th July Evening Shift

$$\underset{\text{Reactant}}{[A]} \rightarrow \underset{\text{Product}}{[B]}$$

If formation of compound $[\text{B}]$ follows the first order of kinetics and after 70 minutes the concentration of $[\text{A}]$ was found to be half of its initial concentration. Then the rate constant of the reaction is $x \times 10^{-6}\ \text{s}^{-1}$. The value of x is _____. (Nearest Integer)

Open solution

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Q198. JEE Main 2022 (Online) 27th July Morning Shift

$2NO + 2H_2 \rightarrow N_2 + 2H_2O$

The above reaction has been studied at $800^\circ C$. The related data are given in the table below

Reaction serial number	Initial Pressure of H_2/kPa	Initial Pressure of NO/kPa	Initial rate $\left(\frac{-dp}{dt}\right)/(kPa/s)$
1	65.6	40.0	0.135
2	65.6	20.1	0.033
3	38.6	65.6	0.214
4	19.2	65.6	0.106

The order of the reaction with respect to NO is _____.

Open solution

Q199. JEE Main 2022 (Online) 26th July Morning Shift

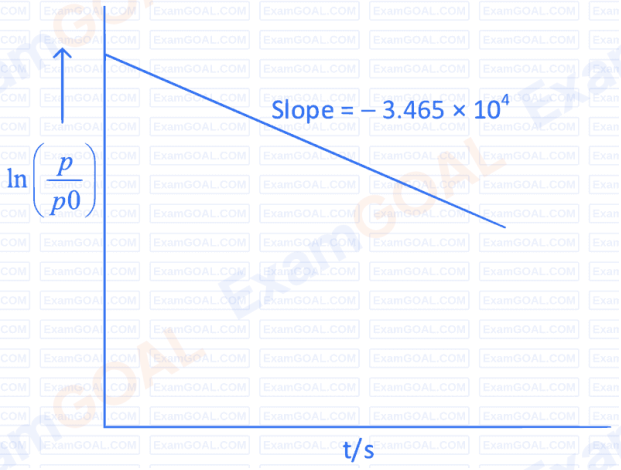
For a reaction $A \rightarrow 2B + C$ the half lives are $100\ s$ and $50\ s$ when the concentration of reactant A is 0.5 and $1.0\ mol\ L^{-1}$ respectively. The order of the reaction is _____ . (Nearest Integer)

Open solution

Q200. JEE Main 2022 (Online) 25th July Evening Shift

For the decomposition of azomethane.

$CH_3N_2CH_3(g) \rightarrow CH_3CH_3(g) + N_2(g)$, a first order reaction, the variation in partial pressure with time at 600 K is given as



The half life of the reaction is _____ $\times 10^{-5}$ s. [Nearest integer]

Open solution

Q201. JEE Main 2022 (Online) 25th July Morning Shift

The half life for the decomposition of gaseous compound A is $240\ s$ when the gaseous pressure was 500 Torr initially. When the pressure was 250 Torr, the half life was found to be 4.0 min. The order of the reaction is _____. (Nearest integer)

Open solution

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Q202. JEE Main 2022 (Online) 30th June Morning Shift

For the reaction P $\rightarrow$ B, the values of frequency factor A and activation energy E_A are 4×10^{13} s^{-1} and 8.3 kJ mol^{-1} respectively. If the reaction is of first order, the temperature at which the rate constant is 2×10^{-6} s^{-1} is _____ $\times 10^{-1}$ K.

(Given : ln 10 = 2.3, R = 8.3 J K^{-1} mol^{-1}, log2 = 0.30)

Open solution

Q203. JEE Main 2022 (Online) 29th June Evening Shift

The equation

$$k = (6.5 \times 10^{12} s^{-1}) e^{-26000K/T}$$

is followed for the decomposition of compound A. The activation energy for the reaction is _____ kJ mol^{-1}. [nearest integer]

(Given : R = 8.314 J K^{-1} mol^{-1})

Open solution

Q204. JEE Main 2022 (Online) 29th June Morning Shift

The activation energy of one of the reactions in a biochemical process is 532611 J mol^{-1}. When the temperature falls from 310 K to 300 K, the change in rate constant observed is $k_{300} = x \times 10^{-3}\ k_{310}$. The value of x is _____.

[Given : $\ln 10 = 2.3$, R = 8.3 J K^{-1} mol^{-1}]

Open solution

Q205. JEE Main 2022 (Online) 28th June Evening Shift

A radioactive element has a half life of 200 days. The percentage of original activity remaining after 83 days is _____. (Nearest integer)

(Given : antilog 0.125 = 1.333, antilog 0.693 = 4.93)

Open solution

Q206. JEE Main 2022 (Online) 28th June Morning Shift

For a first order reaction A $\rightarrow$ B, the rate constant, k = 5.5×10^{-14} s^{-1}. The time required for 67% completion of reaction is $x \times 10^{-1}$ times the half life of reaction. The value of x is _____ (Nearest integer)

(Given : log 3 = 0.4771)

Open solution

Q207. JEE Main 2022 (Online) 27th June Evening Shift

It has been found that for a chemical reaction with rise in temperature by 9 K the rate constant gets doubled. Assuming a reaction to be occurring at 300 K, the value of activation energy is found to be _____ kJ mol^{-1}. [nearest integer]

(Given ln10 = 2.3, R = 8.3 J K^{-1} mol^{-1}, log 2 = 0.30)

Open solution

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Q208. JEE Main 2022 (Online) 27th June Morning Shift

The rate constant for a first order reaction is given by the following equation:

$\ln k = 33.24 - \frac{2.0 \times 10^4\ K}{T}$

The activation energy for the reaction is given by _____ kJ mol^{-1}. (In nearest integer) (Given : R = 8.3 J K^{-1} mol^{-1})

Open solution

Q209. JEE Main 2022 (Online) 26th June Evening Shift

Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300 K. The ratio of rate constants, $\frac{k_{T,\ Catalysed}}{k_{T,\ Uncatalysed}}$ is e^x. The value of x is _____. [nearest integer]

[Assume that the pre-exponential factor is same in both the cases. Given R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q210. JEE Main 2022 (Online) 26th June Morning Shift

A flask is filled with equal moles of A and B. The half lives of A and B are 100 s and 50 s respectively and are independent of the initial concentration. The time required for the concentration of A to be four times that of B is _____ s.

(Given : ln 2 = 0.693)

Open solution

Q211. JEE Main 2022 (Online) 25th June Evening Shift

At 345 K, the half life for the decomposition of a sample of a gaseous compound initially at 55.5 kPa was 340 s. When the pressure was 27.8 kPa, the half life was found to be 170 s. The order of the reaction is _____. [integer answer]

Open solution

Q212. JEE Main 2022 (Online) 25th June Morning Shift

For a given chemical reaction

$\gamma_1 A + \gamma_2 B \rightarrow \gamma_3 C + \gamma_4 D$

Concentration of C changes from 10 mmol dm^{-3} to 20 mmol dm^{-3} in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be 9 mmol dm^{-3} s^{-1}. Therefore, the rate of reaction is _____ mmol dm^{-3} s^{-1}. (Nearest Integer)

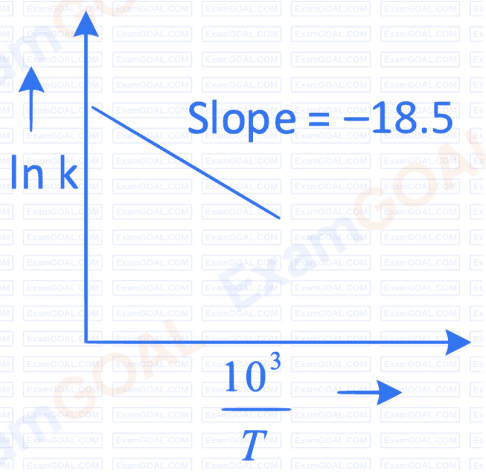
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Q213. JEE Main 2022 (Online) 24th June Morning Shift

The rate constants for decomposition of acetaldehyde have been measured over the temperature range 700 - 1000 K. The data has been analysed by plotting ln k vs $\frac{10^3}{T}$ graph. The value of activation energy for the reaction is _____ kJ mol^{-1}. (Nearest integer)

(Given : R = 8.31 J K^{-1} mol^{-1})



Open solution

Q214. JEE Main 2021 (Online) 31st August Evening Shift

For the reaction A $\rightarrow$ B, the rate constant k(in s^{-1}) is given by

$$\log_{10}k = 20.35 - \frac{(2.47\times10^3)}{T}$$

The energy of activation in kJ mol^{-1} is _____. (Nearest integer) [Given : R = 8.314 J K^{-1} mol $^{-1}$]

Open solution

Q215. JEE Main 2021 (Online) 31st August Morning Shift

For a first order reaction, the ratio of the time for 75% completion of a reaction to the time for 50% completion is _____. (Integer answer)

Open solution

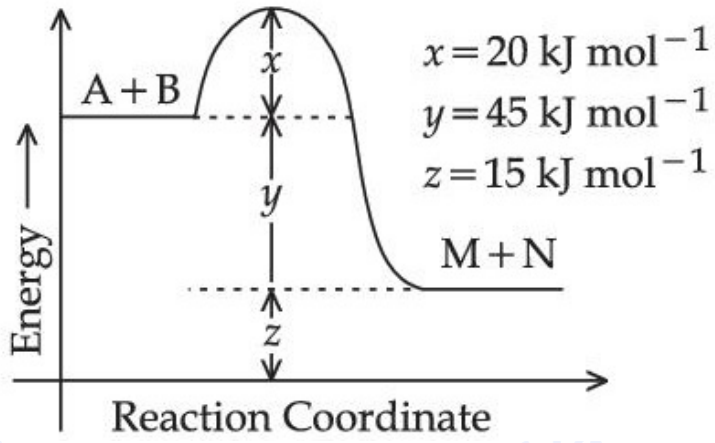
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Q216. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

According to the following figure, the magnitude of the enthalpy change of the reaction

A + B $\rightarrow$ M + N in kJ mol^{-1} is equal to _____. (Integer answer)



Q217. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

The first order rate constant for the decomposition of $CaCO_3$ at 700 K is $6.36 \times 10^{-3} s^{-1}$ and activation energy is 209 kJ mol^{-1}. Its rate constant (in s^{-1}) at 600 K is $x \times 10^{-6}$. The value of x is _____. (Nearest integer)

[Given R = 8.31 J K^{-1} mol^{-1}; log 6.36×10^{-3} = -2.19, $10^{-4.79} = 1.62 \times 10^{-5}$]

Q218. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

The reaction that occurs in a breath analyser, a device used to determine the alcohol level in a person's blood stream is

$$2K_2Cr_2O_7 + 8H_2SO_4 + 3C_2H_6O \rightarrow 2Cr_2(SO_4)_3 + 3C_2H_4O_2 + 2K_2SO_4 + 11H_2O$$

If the rate of appearance of $Cr_2(SO_4)_3$ is 2.67 mol min^{-1} at a particular time, the rate of disappearance of C_2H_6O at the same time is _____ mol min^{-1}. (Nearest integer)

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Q219. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

The following data was obtained for chemical reaction given below at 975 K.

$2NO_{(g)} + 2H_{2(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$

	$[NO]$ $mol\ L^{-1}$	$[H_2]$ $mol\ L^{-1}$	Rate $mol\ L^{-1}s^{-1}$
(A)	8×10^{-5}	8×10^{-5}	7×10^{-9}
(B)	24×10^{-5}	8×10^{-5}	2.1×10^{-8}
(C)	24×10^{-5}	32×10^{-5}	8.4×10^{-8}

The order of the reaction with respect to NO is _____. [Integer answer]

Q220. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

For the first order reaction A $\rightarrow$ 2B, 1 mole of reactant A gives 0.2 moles of B after 100 minutes. The half life of the reaction is _____ min.

[Use : ln 2 = 0.69, ln 10 = 2.3]

Properties of logarithms : $\ln x^y = y \ln x$;

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

(Round off to the nearest integer)

Q221. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

For a chemical reaction A $\rightarrow$ B, it was found that concentration of B is increased by 0.2 mol L^- in 30 min. The average rate of the reaction is _____ $\times 10^{-1}$ mol L^{-1} h^{-1}. (in nearest integer)

Q222. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

$$N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$$

In the above first order reaction the initial concentration of N_2O_5 is 2.40×10^{-2} mol L^{-1} at 318 K. The concentration of N_2O_5 after 1 hour was 1.60×10^{-2} mol L^{-1}. The rate constant of the reaction at 318 K is _____ $\times 10^{-3}$ min^{-1}. (Nearest integer)

[Given : log 3 = 0.477, log 5 = 0.699]

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Q223. JEE Main 2021 (Online) 20th July Evening Shift

$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

In the above first order reaction the concentration of PCl_5 reduces from initial concentration 50 mol L^{-1} to 10 mol L^{-1} in 120 minutes at 300 K. The rate constant for the reaction at 300 K is x $\times$ 10^{-2} min^{-1}. The value of x is _____. [Given log5 = 0.6989]

Open solution

Q224. JEE Main 2021 (Online) 20th July Morning Shift

The inactivation rate of a viral preparation is proportional to the amount of virus. In the first minute after preparation, 10% of the virus is inactivated. The rate constant for viral inactivation is _____ $\times$ 10^{-3} min^{-1}. (Nearest integer)

[Use : ln 10 = 2.303; $\log_{10}$ 3 = 0.477; property of logarithm : log x^y = y log x]

Open solution

Q225. JEE Main 2021 (Online) 18th March Evening Shift

A reaction has a half life of 1 min. The time required for 99.9% completion of the reaction is _____ min. (Round off to the Nearest Integer). [Use : ln 2 = 0.69; ln 10 = 2.3]

Open solution

Q226. JEE Main 2021 (Online) 18th March Morning Shift

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(s)$

This reaction was studied at -10° and the following data was obtained

Run	$[NO]_0$	$[Cl_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall order of the reaction is _____. (Round off to the Nearest Integer).

Open solution

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Q227. JEE Main 2021 (Online) 17th March Evening Shift

The reaction $2A + B_2 \rightarrow 2AB$ is an elementary reaction.

For a certain quantity of reactants, if the volume of the reaction vessel is reduced by a factor of 3, the rate of the reaction increases by a factor of _____. (Round off to the Nearest Integer).

Open solution

Q228. JEE Main 2021 (Online) 17th March Morning Shift

For a certain first order reaction 32% of the reactant is left after 570s. The rate constant of this reaction is _____ $\times 10^{-3}$ s^{-1}. (Round off to the Nearest Integer). [Given : $\log_{10}2 = 0.301$, $\ln 10 = 2.303$]

Open solution

Q229. JEE Main 2021 (Online) 16th March Evening Shift

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is _____ min. (Round off to the Nearest Integer).

Open solution

Q230. JEE Main 2021 (Online) 16th March Morning Shift

The decomposition of formic acid on gold surface follows first order kinetics. If the rate constant at 300 K is 1.0×10^{-3} s^{-1} and the activation energy E_a = 11.488 kJ mol^{-1}, the rate constant at 200 K is _____ $\times 10^{-5}$ s^{-1}. (Round off to the Nearest Integer). (Given : R = 8.314 J mol^{-1} K^{-1})

Open solution

Q231. JEE Main 2021 (Online) 26th February Evening Shift

If the activation energy of a reaction is 80.9 kJ mol^{-1}, the fraction of molecules at 700 K, having enough energy to react to form
products is e^{-x}. The value of x is _____. (Rounded off to the nearest integer) [Use R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q232. JEE Main 2021 (Online) 26th February Morning Shift

An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1}. If energy change ΔE during the reaction is -20 kJ, then the activation energy for the reverse reaction in kJ is _____. (Integer answer)

Open solution

Q233. JEE Main 2021 (Online) 25th February Evening Shift

The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C. The value of activation energy in kJ mol^{-1} is _____. (Rounded off to the nearest integer)

[R = 8.314 J $K^{-1}mol^{-1}$]

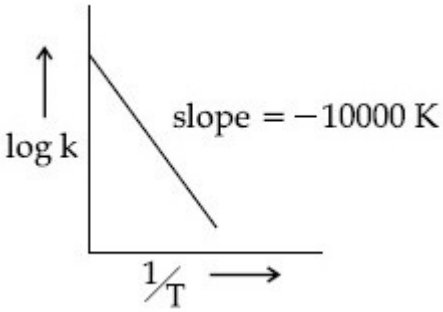
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Q234. JEE Main 2021 (Online) 25th February Morning Shift

For the reaction, aA + bB $\rightarrow$ cC + dD, the plot of log k vs $\frac{1}{T}$ is given below :

Open solution



The temperature at which the rate constant of the reaction is 10^{-4} s^{-1} is _____ K. (Rounded off to the nearest integer)

[Given : The rate constant of the reaction is 10^{-5} s^{-1} at 500 K.]

Q235. JEE Main 2021 (Online) 24th February Evening Shift

Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25°C. After 9 h, the fraction of sucrose remaining is f. The value of $\log_{10}\left(\frac{1}{f}\right)$ is _____ $\times$ 10^{-2}. (Rounded off to the nearest integer)

Open solution

[Assume : ln 10 = 2.303, ln 2 = 0.693]

Q236. JEE Main 2021 (Online) 24th February Morning Shift

Gaseous cyclobutene isomerises to butadiene in a first order process which has a 'k' value of 3.3 $\times$ 10^{-4} s^{-1} at 153°C. The time in minutes it takes for the isomerization to proceed 40% to completion at this temperature is _____.
(Rounded off to the nearest integer)

Open solution

Q237. JEE Main 2020 (Online) 6th September Evening Slot

The rate of a reaction decreased by 3.555 times when the temperature was changed from 40°C to 30°C. The activation energy (in kJ mol^{-1}) of the reaction is _____.

Open solution

Take; R = 8.314 J mol^{-1} K^{-1} ln 3.555 = 1.268

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Q238. JEE Main 2020 (Online) 4th September Evening Slot

The number of molecules with energy greater than the threshold energy for a reaction increases five fold by a rise of temperature from 27°C to 42°C. Its energy of activation in J/mol is _____.
(Take ln 5 = 1.6094; R = 8.314 J mol^{-1} K^{-1})

Open solution

Q239. JEE Main 2020 (Online) 4th September Morning Slot

If 75% of a first order reaction was completed in 90 minutes, 60% of the same reaction would be completed in approximately (in minutes) _____.

(Take : log 2 = 0.30; log 2.5 = 0.40)

Open solution

Q240. JEE Main 2020 (Online) 9th January Evening Slot

A sample of milk splits after 60 min. at 300 K and after 40 min. at 400 K when the population of lactobacillus acidophilus in it doubles. The activation energy (in kJ/ mol) for this process is closest to_____.

(Given, R = 8.3 J mol^{-1} K^{-1}, $\ln\left(\frac{3}{2}\right) = 0.4$, e^{-3} = 4.0)

Open solution

Q241. JEE Main 2020 (Online) 7th January Morning Slot

During the nuclear explosion, one of the products is ^{90}Sr with half life of 6.93 years. If 1 μ g of ^{90}Sr was absorbed in the bones of newly born baby in place of Ca, how much time, in years, is required to reduce it by 90% if it is not lost metabolically?

Open solution

Out of syllabus 2 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q242. JEE Main 2025 (Online) 22nd January Morning Shift · MCQ (Single Correct Answer)

Which of the following statement is not true for radioactive decay?

(A) Half life is $\ln 2$ times of $\frac{1}{\text{rate constant}}$.

(B) Decay constant increases with increase in temperature.

(C) Decay constant does not depend upon temperature.

(D) Amount of radioactive substance remained after three half lives is $\frac{1}{8}$ th of original amount.

Open solution

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Q243. JEE Main 2023 (Online) 6th April Morning Shift · **Numerical**

Open solution

For the adsorption of hydrogen on platinum, the activation energy is $30\ \mathrm{k\ Jmol^{-1}}$ and for the adsorption of hydrogen on nickel, the activation energy is $41.4\ \mathrm{k\ Jmol^{-1}}$. The logarithm of the ratio of the rates of chemisorption on equal areas of the metals at $300\ \mathrm{K}$ is _____ (Nearest integer)

Given: $\ln 10 = 2.3$

$\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **B**	4. **D**	5. **A**
6. **B**	7. **B**	8. **D**	9. **A**	10. **D**
11. **A**	12. **C**	13. **D**	14. **B**	15. **D**
16. **B**	17. **C**	18. **D**	19. **A**	20. **B**
21. **A**	22. **B**	23. **C**	24. **B**	25. **D**
26. **C**	27. **C**	28. **B**	29. **C**	30. **B**
31. **B**	32. **C**	33. **B**	34. **D**	35. **D**
36. **A**	37. **C**	38. **D**	39. **B**	40. **A**
41. **A**	42. **C**	43. **C**	44. **A**	45. **C**
46. **B**	47. **A**	48. **B**	49. **D**	50. **D**
51. **B**	52. **A**	53. **D**	54. **A**	55. **B**
56. **B**	57. **D**	58. **B**	59. **B**	60. **D**
61. **D**	62. **B**	63. **B**	64. **A**	65. **A**
66. **A**	67. **A**	68. **A**	69. **D**	70. **B**
71. **B**	72. **B**	73. **B**	74. **B**	75. **C**
76. **D**	77. **A**	78. **B**	79. **C**	80. **A**
81. **D**	82. **A**	83. **C**	84. **C**	85. **A**
86. **D**	87. **D**	88. **B**	89. **D**	90. **B**
91. **C**	92. **B**	93. **C**	94. **D**	95. **B**
96. **B**	97. **D**	98. **C**	99. **B**	100. **D**
101. **D**	102. **B**	103. **B**	104. **C**	105. **D**
106. **C**	107. **B**	108. **D**	109. **A**	110. **D**
111. **A**	112. **C**	113. **A**	114. **A**	115. **D**
116. **A**	117. **C**	118. **A**	119. **C**	120. **B**
121. **D**	122. **A**	123. **B**	124. **B**	125. **C**
126. **B**	127. **D**	128. **B**	129. **C**	130. **B**
131. **A**	132. **A**	133. **D**	134. **D**	135. **A**
136. **C**				

Numerical

137. **232**	138. **25**	139. **90**	140. **46**	141. **7**

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142. **20**	143. **41**	144. **5**	145. **102**	146. **10**
147. **57**	148. **1303**	149. **8**	150. **43**	151. **2435**
152. **20**	153. **897**	154. **69**	155. **3**	156. **8**
157. **1**	158. **17**	159. **3**	160. **315**	161. **3**
162. **50**	163. **100**	164. **2**	165. **17190**	166. **399**
167. **17**	168. **24**	169. **63**	170. **30**	171. **10**
172. **2**	173. **130**	174. **2200**	175. **3**	176. **3**
177. **333**	178. **1**	179. **2**	180. **2**	181. **75**
182. **15**	183. **17**	184. **2520**	185. **1350**	186. **623**
187. **2**	188. **1**	189. **1**	190. **60**	191. **3**
192. **3**	193. **1**	194. **40**	195. **8**	196. **100**
197. **165**	198. **2**	199. **2**	200. **2**	201. **1**
202. **225**	203. **216**	204. **1**	205. **75**	206. **16**
207. **59**	208. **166**	209. **4**	210. **200**	211. **0**
212. **1**	213. **154**	214. **47**	215. **2**	216. **45**
217. **16**	218. **4**	219. **1**	220. **600 to 700**	221. **4**
222. **7**	223. **1**	224. **106**	225. **10**	226. **3**
227. **27**	228. **2**	229. **108**	230. **10**	231. **14**
232. **50**	233. **52**	234. **526**	235. **81**	236. **26**
237. **100**	238. **84297.47 to 84297.48**	239. **60**	240. **3.98 to 3.99**	241. **23 to 23.03**

Out of syllabus

242. **B**	243. **2**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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